

Collateral Policy Surprises*

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Abstract

Central bank collateral policy specifies which assets banks can pledge as collateral to obtain central bank funding and is an important determinant of liquidity in the banking system. We propose a high-frequency identification approach to study the effects of central bank collateral policy on banks and the government bond market. We identify collateral policy surprises using intraday bank stock price changes around Eurosystem collateral policy announcements. Collateral easing induces excess returns of bank stocks and reduces bank credit default swap (CDS) spreads, in particular for riskier banks and banks located in the euro area periphery. Furthermore, collateral easing compresses core-periphery government bond and sovereign CDS spreads. The effects are stronger for policy changes that affect the collateral treatment of government bonds or bank debt, but still significant for unrelated events. The uneven transmission of collateral policy through banks to sovereign bond markets is distinct from both central bank asset purchases and conventional monetary policy.

Keywords: Central Bank Collateral Framework, Bank Liquidity Risk, Government Bond Market, High Frequency Identification, Intermediary Asset Pricing

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1 Introduction

Central banks implement monetary policy by lending to banks against collateral. In times of financial stress, central banks typically expand their collateral frameworks, either by accepting additional assets or lowering valuation haircuts. By easing banks' access to central bank funding, such measures are commonly viewed as crisis-management tools that support financial stability and monetary policy transmission. In this paper, we document that this crisis-centered view of collateral policy is incomplete: collateral policy can meaningfully shape financial markets at any time. Using a novel high-frequency identification approach, we show that collateral policy systematically affects banks' equity values and default risk, as well as the pricing of assets pledged as collateral, such as government bonds. To quantify these effects and identify the transmission channels at play, we exploit changes to the Eurosystem collateral framework as a laboratory.

Establishing causal effects of central bank collateral policy on banks and financial markets is challenging for two main reasons. First, the Eurosystem accepts a wide range of assets as collateral, some of which are illiquid, and policy announcements often affect multiple asset classes simultaneously. This makes the choice of an appropriate asset class to study the systematic effects of collateral policy nontrivial. Second, the stance of collateral policy varies over time and endogenously responds to the macro-financial environment. For example, during the Great Financial Crisis in 2008, the European Central Bank (ECB) and several other central banks reduced the minimum rating requirements on government and corporate bonds to ensure banks' access to central bank funding and safeguard monetary policy transmission. Thus, it is difficult to separate the systematic from the surprise (unanticipated) component of a collateral policy change, which is essential for establishing causal effects.

We use a stylized intermediary asset pricing model to overcome the first challenge and to guide our choice of an appropriate asset class for measuring collateral policy surprises. In the model, banks receive liquidity shocks that they have to settle by borrowing from the central bank against collateral, in the spirit of Bianchi and Bigio (2022) and De Fiore et al. (2024). A relaxation of collateral policy reduces the probability that banks' liquidity shocks exceed the collateral value of their assets. The stylized model predicts that expansionary collateral policy increases bank stock prices and reduces bank credit default swap (CDS) spreads, in particular for riskier banks and those with a smaller share of eligible assets.

Motivated by the model, we propose a novel identification strategy based on high-frequency bank stock price reactions around Eurosystem collateral policy announcements. Focusing on banks as the users of collateral rather than the eligible assets themselves has several advantages. It allows us to include the broadest possible set of collateral policy events. In particular, we can include events concerning assets for which market prices are not available. We avoid many judgment calls on the choice of assets when multiple asset classes are affected by a policy change. In addition, extracting the common factor from individual responses does not require information on individual banks' exposure to targeted assets, which cannot be measured at sufficiently high frequency, even with confidential securities holdings data.

Concerning the second challenge to identification, the high-frequency approach enables us to separate the surprise component of collateral policy from systematic policy responses to macro-financial conditions. Our approach is conceptually similar to the identification of monetary policy shocks using high-frequency asset price data (Kuttner, 2001, Gürkaynak et al., 2005), which has recently been applied to other macro-financial policies, such as bank capital regulation (Drechsel and Miura, 2026). Lastly, our high-frequency identification strategy captures expected general equilibrium effects of collateral policy changes, rather than only their direct effects on targeted assets or banks.

We implement this market-based identification strategy in two steps: we identify relevant collateral policy announcements and measure the common intraday response of euro area bank stocks around each event. We compile a comprehensive list of ECB collateral policy announcements from the introduction of a harmonized Eurosystem collateral framework in January 2007 to December 2022, building on Bindseil et al. (2017). We find 55 collateral policy events that attracted sufficient attention in financial markets to be reported by Thomson Reuters News. Excluding announcements that coincide with ECB monetary policy decision dates leaves a final sample of 43 events. We compute stock price changes of the largest euro area banks over a narrow intraday (45-minute) window around these announcements and extract the first principal component to isolate the common factor in bank stock reactions. These intraday price reactions are often sizable, exceeding two percent for some announcements. Furthermore, bank stock returns are considerably more volatile on event dates than non-event dates. The magnitude of these reactions is also economically meaningful, with bank stocks moving by around one percent in response to collateral policy surprises, compared with 0.4 percent in response to conventional monetary policy surprises. We interpret positive (negative) reactions as expansionary (contractionary) collateral policy surprises.

The identified high-frequency collateral policy surprise series exhibits no significant autocorrelation and is not predictable based on information available to financial markets prior to the announcements (Bauer and Swanson, 2023). We assess predictability using a broad set of macroeconomic and financial variables, as well as ECB monetary policy surprises (Altavilla et al., 2019), using various alternative specifications and forecast horizons. Across these tests, we find no systematic evidence that the collateral policy surprise series is predictable. In addition, we verify that our event windows do not contain potentially confounding macroeconomic data releases (Gürkaynak et al., 2020).

Furthermore, we do not find significant evidence for central bank information effects (Jarociński and Karadi, 2020; Miranda-Agrippino and Ricco, 2021), as bank stocks and CDS spreads move in opposite directions for the majority of events, including all events associated with large bank stock price reactions. Moreover, we show that bank stock prices are much more volatile on days with a collateral policy event than on control days without such announcements, indicating that these announcements trigger significant market reactions (Känzig, 2021). Lastly, the market-based classification into expansionary and contractionary events generally aligns with narrative classification of events. To further validate our empirical approach, we demonstrate that illiquid

and weakly capitalized banks respond more strongly to changes in collateral policy. This finding is in line with our model and with a large body of literature on the bank lending channel of conventional monetary policy (Kashyap and Stein, 2000), further corroborating the validity of the high-frequency collateral policy surprise series.

We next use the collateral policy surprise (CPS) series to examine the effects of collateral policy on financial markets. Using local projections (Jordà, 2005) estimated on daily data, we find that expansionary collateral policy generates broad-based equity gains, with bank stocks significantly outperforming the broader market. Collateral easing also lowers the most commonly traded equity market volatility index (V2X), suggesting that it dampens financial market uncertainty. In addition, collateral easing lowers bank CDS spreads for all banks, consistent with empirical evidence in Legroux et al. (2022), who use the difference between ECB and private repo market haircuts to measure the effects of central bank liquidity support. We further find that the decline in CDS spreads is substantially larger for banks located in euro area periphery countries, exceeding 10 basis points for some institutions, compared with less than 5 basis points for banks in core countries.

We then examine how expansionary collateral policy affects the government bond market. The spreads of all major euro area sovereigns over the overnight index swap (OIS) decline significantly following an expansionary collateral policy surprise. The decline is around 10 basis points for periphery bonds (Italy and Spain) but only about 2 basis points for core bonds (France and Germany). Accordingly, the periphery-core government bond spread narrows substantially. Collateral policy also affects the pricing of sovereign risk more generally. We observe a decline in sovereign credit default swap (CDS) spreads, which is again much stronger for periphery borrowers. While events related to the collateral treatment of government bonds induce larger responses, we still observe significant and economically relevant effects for events unrelated to the collateral treatment of government bonds. We find no significant response of the CDS-bond basis, which is a common measure of bond convenience yields and defined as the difference between the government bond yield and a replication portfolio of CDS-spread and the risk-free rate.

To reconcile the heterogeneous transmission of collateral policy to the government bond market and the absence of effects on the CDS-bond basis, we revisit our stylized intermediary asset pricing framework. The model predicts that lower bank default risk raises government bond prices, since banks receive the payoff from their investment with a larger probability. This effect is more pronounced for riskier banks, which is exactly what we find in the data. Importantly, the most responsive banks are located in periphery countries, exhibit higher CDS spreads, lower balance sheet liquidity, poorer capitalization, and a larger home bias in sovereign bond holdings.

We argue that home bias is a key determinant of the uneven transmission of collateral policy.¹ Following an expansionary collateral policy surprise, periphery banks exhibit a larger drop in their CDS spreads than core banks. Since periphery banks are the main holders of

¹The home bias of European banks' holdings of domestic sovereign debt and its adverse consequences are documented in Acharya and Steffen (2015) and Ongena et al. (2019).

periphery government bonds, these bonds also exhibit the largest yield response. This novel transmission mechanism of collateral policy through bank default risk can help explain why periphery bond and CDS spreads are more responsive to collateral policy, while the CDS-bond basis does not respond for any country. Consistent with this mechanism, we document that the effects of collateral policy on banks and government bonds are strongest for events related to the sovereign-bank nexus. Such events include minimum rating requirement exemptions for Greece in 2012 and 2015 and changes in the eligibility rules of unsecured bank debt.

Moreover, we shed light on the roles of the business cycle and banks' excess liquidity in the transmission of collateral policy. We find evidence that the effects of collateral policy are particularly pronounced during recessions, when banks' demand for central bank liquidity is likely to be greater than during expansions. By contrast, during periods of excess liquidity, especially following the introduction of the Eurosystem large-scale asset purchases in 2015, the effects of collateral policy are less pronounced, but not zero.

Lastly, we benchmark the effects of collateral policy surprises against high-frequency surprises associated with other central bank instruments, including conventional interest rate policy, forward guidance and quantitative easing (Altavilla et al., 2019). None of these other policy dimensions generates significant excess returns for bank stocks, nor do we find comparable heterogeneous effects on government bond yields or on bank and sovereign CDS spreads. These findings point to a markedly different transmission mechanism of collateral policy to banks and asset prices than that of monetary policy.

Related Literature Our paper contributes to two strands of literature. First, from a methodological perspective, it builds on the literature on high-frequency identification. The key insight of this literature is that the unexpected (surprise) component of policy announcements can be identified using asset price movements within narrow time windows surrounding these announcements, an approach that has been used extensively in the empirical monetary policy literature (Kuttner, 2001, Gürkaynak et al., 2005, Gertler and Karadi, 2015, Altavilla et al., 2019, Swanson, 2021, Miranda-Agrippino and Ricco, 2021).² More broadly, this approach has been extended beyond monetary policy to study the transmission of other macro-financial policies. Boneva et al. (2023) analyze FOMC announcements related to lender-of-last-resort policy, and Bluwstein and Patozi (2024) examine macroprudential policy announcements. Using a similar methodological setup, Phillot (2025) identifies Treasury supply shocks in bond markets, while Diebold and Hack (2026) study sovereign credit risk shocks. Furthermore, Drechsel and Miura (2026) use bank stock reactions around Federal Reserve speeches to identify news about bank regulation and estimate its macroeconomic effects. Ottonello and Song (2025) use changes in financial intermediaries' stock prices around earnings announcements to identify shocks to intermediary net worth and trace their effects on non-financial firms. We construct a novel high-frequency identification approach to Eurosystem collateral policy, using the common intraday response of

²Fecht et al. (2026) exploit transaction-level data at the bank level to measure the transmission of bank-specific conventional monetary policy surprises.

bank stock prices around collateral policy announcements to construct a collateral policy surprise series. To the best of our knowledge, this is the first high-frequency measure of collateral policy surprises.

Second, our paper relates to the empirical banking literature on central bank collateral policy, which typically focuses on individual policy changes and exploits security- or loan-level variation. This literature shows that central bank collateral frameworks have economically meaningful effects on banks (the users of collateral) as well as on non-financial firms and sovereigns (the issuers of collateral). Expansionary collateral policy reduces bank funding costs (Cassola and Koulischer, 2019; Fang et al., 2025), affects the prices of eligible assets (Mésonnier et al., 2021; Chen et al., 2023), increases bank credit supply (Bekkum et al., 2018), and has real effects on the non-financial sector (Delatte et al., 2025; Hüttl and Kaldorf, 2026). Jasova et al. (2024) find that banks pledge and hold more bonds with a large haircut gap between the Eurosystem collateral framework and the private repo market and document that this leads to higher issuance of such bonds. Pelizzon et al. (2024) show that the inclusion of corporate bonds as eligible collateral lowers yields and improves secondary market liquidity. For sovereign bonds, Nissinen and Sihvonen (2024) document that yields respond to discrete changes in collateral haircuts across maturity buckets, while Nguyen (2020) shows that investors demand higher yields for bonds with higher collateral haircuts. Legroux et al. (2022) demonstrate that lenient ECB haircut policies have a benign effect on banks’ liquidity risk. We contribute to this literature by providing a systematic analysis of collateral policy’s macro-financial effects using the full set of policy changes rather than relying on individual events. Importantly, several findings from our high-frequency approach resonate with the key findings documented in the micro-level literature.

Outline The remainder of the paper is structured as follows. Section 2 reviews the Eurosystem collateral framework and introduces a stylized intermediary asset pricing model to illustrate our identification strategy. We describe our data and measurement, and provide a series of diagnostic tests on the identified surprise series in Section 3. In Section 4, we discuss the effects of collateral policy surprises on financial variables and on the government bond market. Section 5 concludes.

2 Collateral Policy: Institutional Background and Theory

This section provides the institutional background of the Eurosystem collateral framework underlying our empirical results. Furthermore, we introduce a highly stylized model of collateral policy and banks, which delivers predictions of the relationship between collateral policy, bank stock returns, and the government bond market. Guided by the institutional background and the model, we then describe our identification strategy and its advantages.

2.1 Institutional Background

Central banks implement monetary policy by lending to commercial banks against collateral. Unlike the interbank market, there is no unsecured funding option. A sufficiently large pool

of eligible collateral is therefore necessary to give all euro area banks access to central bank facilities and to ensure the smooth implementation of monetary policy. The Eurosystem collateral framework is exceptionally broad. While the Federal Reserve lends predominantly against highly liquid US Treasury bonds, euro area banks can pledge a wide variety of asset classes, including non-marketable assets and BBB-rated bonds, see Bindseil et al., 2017 for an overview. The main reason for operating such a broad framework is that the amount of euro-denominated high-quality liquid assets, such as AAA-rated government bonds, is small relative to the size and regional heterogeneity of the euro area banking sector. Establishing the causal effects of collateral policy on asset prices is therefore challenging: announcements can concern a variety of asset classes, and may affect multiple asset classes simultaneously. Furthermore, collateral policy itself operates along two distinct dimensions: eligibility requirements and valuation haircuts.

Additionally, when accepting financial assets as collateral, the Eurosystem sets minimum quality requirements on assets and applies valuation haircuts to the market value of the pledged asset.³ This reduces losses from exposure to low-quality collateral in the case of a counterparty default. Accepting risky collateral can be costly either because setting up and operating a credit risk management facility is costly for the central bank (Bindseil and Papadia, 2006; Hall and Reis, 2015), or because central banks are loss-averse (Goncharov et al., 2023). Therefore, the Eurosystem sets its collateral framework sufficiently wide to facilitate a smooth implementation of monetary policy, but tight enough to limit expected losses to an acceptable level.

The macro-financial environment shapes how this trade-off is optimally resolved. In times of financial market stress, banks' demand for central bank funding rises while the availability of high-quality collateral typically declines, as rating downgrades affect issuers in both the private and public sectors. It is then generally optimal for the central bank to expand its collateral framework, both by easing minimum eligibility requirements and by reducing valuation haircuts. Such a systematic reaction of collateral policy complicates identification, since asset prices already reflect anticipated policy responses. For instance, an easing of minimum rating requirements may have contractionary price effects if market participants had expected an even larger relaxation during a period of elevated stress. A careful empirical investigation must therefore account for this endogeneity by employing a suitable instrument for collateral availability.

2.2 A Stylized Model of Collateral Policy

Based on the preceding considerations, the collateral framework is essentially a function that maps the characteristics of bank assets (asset class, default risk, liquidity, size) into their collateral value. Hence, banks appear to be a well-suited candidate to measure collateral policy, as they are the direct users of central bank facilities and the primary channel through which collateral policy transmits to financial markets. To illustrate the transmission mechanisms at play and to guide our identification strategy, we present a stylized intermediary asset pricing

³Importantly, collateral eligibility requirements and valuation haircuts in the Eurosystem only depend on the pledged collateral and not on the borrowing bank. This means that the collateral framework cannot be used to discriminate directly between banks.

model (He and Krishnamurthy, 2013). Earlier models of central bank operational frameworks, such as Repullo (2005) or Hauck and Neyer (2014), focused on the allocation of liquidity across banks and placed less emphasis on the pricing of eligible collateral.

There are two periods indexed by $t = 0, 1$. The risk-free rate is normalized to zero. Banks raise funds by issuing debt to investors d_1 at price q_0^D and invest into fixed income securities b_1 , such as government bonds, corporate bonds, and non-marketable assets, including bank loans to the corporate sector. Their price is denoted by q_0^B , and their expected per-unit payoff is denoted by $\mathcal{R}_1$. Banks cannot raise outside equity in period 0, which translates into a non-negativity constraint on div_0 , and they maximize expected dividends in the next period, $\mathbb{E}_0[div_1]$. Dividends in period 0 are given by:

$$div_0 = q_0^D d_1 - q_0^B b_1 . \quad (1)$$

Since banks only value dividend payouts in period 1, they will not raise more debt than necessary to finance their investment, such that $div_0 = 0$ and equation (1) essentially becomes a balance sheet constraint. At the beginning of period 1, banks draw a liquidity deficit $\delta_1 \cdot d_1$, where δ_1 is a random variable with pdf $f(\delta_1)$ and cdf $F(\delta_1)$, in the spirit of Bianchi and Bigio (2022) and De Fiore et al. (2024). We do not impose a specific distributional assumption, but merely require that the pdf is decreasing over δ_1 , i.e., $f'(\delta_1) < 0$. For instance, the exponential distribution satisfies this assumption. Intuitively, it is increasingly unlikely to draw higher liquidity deficits. Banks settle their liquidity with the central bank against eligible collateral. The collateral value of bank assets depends on the central bank's collateral policy:

$$\bar{b}_1 = (1 - \kappa) \mathcal{R}_1 b_1 , \quad (2)$$

where $\kappa \in [0, 1]$ is the central bank collateral parameter. In this simplified setting, κ should be interpreted as the stance of collateral policy, which encompasses both eligibility criteria and valuation haircuts. Specifically, κ reflects the weighted average haircut on all bank assets, where $\kappa = 1$ corresponds to a 100% haircut on ineligible assets. We assume that banks default if their collateral is insufficient to replace the withdrawn funds by central bank borrowing, i.e., if $\bar{b}_1 < \delta_1 \cdot d_1$.⁴

Bank Stocks and Bank CDS We obtain the bank default probability $D^{bank}(\bar{b}_1/d_1)$ from evaluating the cdf of the liquidity deficit at $\bar{b}_1/d_1$:

$$D^{bank}(\bar{b}_1/d_1) = 1 - F(\bar{b}_1/d_1) . \quad (3)$$

⁴Ashcraft et al. (2010) study central bank haircut policies in a model where banks face collateral constraints. Our model delivers similar predictions and also allows us to link central bank collateral policy directly to bank CDS spreads.

Importantly, the bank default probability can be measured by bank CDS spreads in the data. All else equal, removing the collateral eligibility of specific asset classes or raising the applicable collateral haircut increases the bank failure probability, which corresponds to raising κ in the model. This directly follows from differentiating the default probability with respect to κ :

$$\frac{\partial D^{bank}(\bar{b}_1/d_1)}{\partial \kappa} = -\frac{\partial \bar{b}_1}{\partial \kappa} \frac{1}{d_1} f(\bar{b}_1/d_1) > 0. \quad (4)$$

The first partial derivative $\frac{\partial \bar{b}_1}{\partial \kappa} < 0$ measures how much a change in collateral policy changes collateral availability. If a collateral policy change only affects one asset class, banks that hold a large share of such assets are naturally more affected. If a collateral policy change affects the collateral value of all assets homogeneously, we can write $\frac{\partial \bar{b}_1}{\partial \kappa} = -\mathcal{R}_1 b_1$. The second term measures how much a change in collateral availability affects the default risk, which corresponds to the pdf of the liquidity shock distribution.

While the exposure of an individual bank to a specific collateral policy change is very hard to measure accurately, even with confidential securities holdings data, the model also offers some predictions that can be tested with publicly available data. Specifically, the effect of a homogeneous collateral policy change is more pronounced for riskier banks. Holding the share of eligible assets $\bar{b}_1/d_1$ fixed, this is the case for banks with a more dispersed liquidity shock distribution and, hence, a larger $f(\bar{b}_1/d_1)$. Furthermore, holding the liquidity distribution fixed, the effects are larger for banks with a smaller share of eligible assets $\bar{b}_1/d_1$. This could either be the case if a bank is more levered b_1/d_1 than another bank with the same collateral value $(1 - \kappa)$ of their assets; if a bank has a lower collateral value $(1 - \kappa)$ than a bank with the same leverage ratio; or both.⁵

To link collateral policy to bank stock prices s_0 , we consider expected dividends in period 1. Dividends in period 1 depend on bank survival $div_1 = \mathbb{1}\{\bar{b}_1 > \delta_1 d_1\}(\mathcal{R}_1 b_1 - d_1)$, such that we can write expected dividends as

$$s_0 \equiv \mathbb{E}_0[div_1] = F(\bar{b}_1/d_1)(\mathcal{R}_1 b_1 - d_1). \quad (5)$$

It immediately follows that stock prices are negatively related to κ , i.e., they fall if the collateral haircut parameter κ rises.

$$\frac{\partial s_0}{\partial \kappa} = \frac{\partial \bar{b}_1}{\partial \kappa} f(\bar{b}_1/d_1)(\mathcal{R}_1 b_1 - d_1) < 0. \quad (6)$$

Again, a homogeneous collateral policy tightening (κ rises) has a more pronounced effect on riskier banks' stock prices (high $f(\cdot)$), or on those banks with a smaller share of eligible assets (small $\bar{b}_1/d_1$).

⁵If we differentiate equation (A.5) with respect to the share of eligible assets $\bar{b}_1/d_1$, the monotonicity assumption on the liquidity shock distribution directly implies that $\frac{\partial^2 D^{bank}(\bar{b}_1/d_1)}{\partial \kappa \partial (\bar{b}_1/d_1)} = \frac{\mathcal{R}_1 b_1}{d_1} f'(\bar{b}_1/d_1) < 0$.

Model Implication Ia. Expansionary collateral policy increases bank stock prices and reduces bank CDS spreads.

Model Implication Ib. The effects are stronger for banks with poor capitalization, higher liquidity risk, and a smaller share of eligible assets.

2.3 Identifying Collateral Policy Surprises

Guided by the implications of our simple model, we use bank stock returns around collateral policy announcements to identify collateral policy surprises.

Several advantages arise from focusing on banks as the users of collateral, rather than the issuers of collateral. First, it allows us to examine the largest possible set of collateral policy events, including those events concerning non-marketable assets, such as asset-backed securities or credit claims, for which market prices are not available, at least not at a sufficiently high frequency. Second, it avoids judgment calls on the choice of which asset price to examine whenever an event affects multiple asset classes simultaneously. Third, extracting the common factor from individual bank stock price responses does not require information on the exposure of individual banks to targeted assets, which are often very opaque, even with detailed securities holdings data. Fourth, bank stocks of all major European banks trade at very high frequency on centralized, liquid markets, whereas bank CDS spreads are less liquid and not available for all large euro area banks.

By employing high-frequency bank stock price data within a narrow window around collateral policy announcements, we isolate the unexpected component of the policy change. In doing so, we exclude the systematic responses of the central bank to macro-financial conditions, which is essential for a causal interpretation of the effect of collateral policy surprises on the macroeconomy. Our approach is conceptually similar to the identification of monetary policy shocks using high-frequency asset price data (Kuttner, 2001), which has recently been applied to other macro-financial policies, such as bank capital regulation (Drechsel and Miura, 2026). Finally, an identification strategy based on market prices fully reflects the general equilibrium effects of collateral policy changes.

We interpret a collateral policy event with a stock price increase (decline) as expansionary (contractionary). This classification is purely driven by market data, which comes at a low informational burden about the technical details of each announcement and renders events affecting very different asset classes comparable.

3 Measuring Collateral Policy Surprises

To measure collateral policy surprises, we use high-frequency bank stock price changes in a short window around collateral policy changes.

3.1 Data

Our measurement relies on three data sources. These are related to bank stock prices, the Eurosystem collateral framework for the list of policy announcements and press releases, and historical news archives, which provide additional information on the events.

Bank Stock Prices We obtain high-frequency tick-by-tick data of stock prices of the largest euro area banks contained in the EURO STOXX Banks Index from Thomson Reuters. We have full coverage for BBVA, Caixa Bank, and Santander (Spain); Banco Comercial Portugues (Portugal); BNP Paribas and Societe Generale (France); Intensa Sanpaolo and Unicredit (Italy); and Deutsche Bank (Germany) for our sample period, from 1 January 2007 to 31 December 2022. For ING Group (Netherlands) and Commerzbank (Germany), the high-frequency data are only available for a more recent sub-sample.

ECB Collateral Policy Announcements We obtain a comprehensive list of collateral policy announcements by the ECB since the establishment of the Eurosystem-wide single list in January 2007 until December 2022, extending the list of framework changes contained in Bindseil et al. (2017). We do not include announcements concerning the so-called “Additional Credit Claims” programs by different national central banks during the Great Financial Crisis. Over this period, we identify 98 distinct collateral-related announcements - far from being an exceptionally rigid part of the plumbing of the financial system, collateral policy undergoes six changes per year on average.

Collateral policy announcements are spread across the year and do not systematically coincide with the regular governing council meetings. Unlike monetary policy announcements, which are tied to regular meetings by the ECB governing council, collateral policy changes are often announced in a standalone press release. As an illustration, consider the following excerpt from a press release concerning a collateral policy change on June 22, 2012:

PRESS RELEASE, 22 June 2012 ECB takes further measures to increase collateral availability for counterparties

*On 20 June 2012 the Governing Council of the European Central Bank (ECB) decided on additional measures to improve the access of the banking sector to Eurosystem operations in order to further support the provision of credit to households and non-financial corporations. The Governing Council has reduced the rating threshold and amended the eligibility requirements for certain asset-backed securities (ABSs). It has thus broadened the scope of the measures to increase collateral availability [...]*⁶

Historical News Archive Based on the list of ECB collateral announcements, we make use of the LSEG Historical News Archive to uncover the precise time when news about ECB

⁶The full announcement is accessible under <https://www.ecb.europa.eu/press/pr/date/2012/html/pr120622.en.html>

collateral policy announcements is disseminated to financial market participants. While ECB press releases provide the publication date, they do not include a precise timestamp, making the news archive essential for constructing accurate event windows in our high-frequency analysis.

In addition to timing, the archive provides valuable qualitative insights into market perceptions and commentary surrounding each announcement. For example, following up on the press release 22 June 2012 shown in the previous section, the market reaction for this announcement was the following: *“It just means that they are willing to take as collateral lower-quality credit, which is probably why the periphery is getting a bit of a bid against Germany”*, a trader said. This suggests that markets pay close attention and asset prices responded accordingly; the government bond and stock markets reacted positively.

We restrict our sample to events that received press coverage in the LSEG Historical News Archive, excluding announcements without any headline as insufficiently salient to financial market participants. This gives us 54 events, a complete list of which can be found in Appendix Appendix A.1. We further classify events into those affecting the collateral treatment of government bonds and those exclusively affecting the collateral treatment of privately issued assets. Out of 54 events, 20 are sovereign-related. Finally, we exclude collateral policy announcements coinciding with ECB monetary policy decisions, as well as the EU summit on 28 June 2012, since on these days price movements in the narrow event window cannot be cleanly attributed to the collateral policy change alone. This leaves a final sample of 43 collateral policy announcement events.

3.2 Collateral Policy Surprise Series

By using high-frequency asset price data around collateral policy announcements, we isolate the surprise component of each announcement. Akin to the large literature that identifies monetary policy surprises with high-frequency financial data (Kuttner, 2001; Gürkaynak et al., 2005), we assume that bank stock prices already reflect all relevant information about the macro-financial environment, in particular systematic changes in collateral policy. Thus, any immediate price change in a narrow window around an announcement reflects new information.

We measure the surprise component of each announcement using the bank stock price changes of the largest banks in the Euro STOXX Banks Index in a 45-minute time window around the announcements. Since our focus is on aggregate effects of collateral policy, we extract the first principal component of individual bank stock price reactions, which essentially absorbs all bank-specific characteristics. We will revisit the role of bank heterogeneity in the next section. All events with a positive stock price reaction are interpreted as “expansionary”, i.e., the collateral framework was relaxed more aggressively or tightened less aggressively than expected by market participants.

Figure 1 graphically shows the time series of our collateral policy surprises (CPS). While there are more events during times of high financial stress, in particular during the European sovereign debt crisis, each year contains at least one event. Notably, the magnitude of the bank

stock price reaction is quite evenly distributed across the sample, suggesting that collateral policy has relevant effects on banks outside of crisis times. There are several large positive and large negative events; the shock series has no visible skewness, there is no significant autocorrelation, and we find no heteroskedasticity in the surprise series. Red dots indicate events affecting the collateral treatment of government bonds. Importantly, these events are not clustered around the euro area debt crisis. By contrast, orange events indicate events that exclusively affect the collateral treatment of assets issued by private agents. Some of these events also induced large bank stock price reactions. We run a battery of robustness checks in Section 3.4.

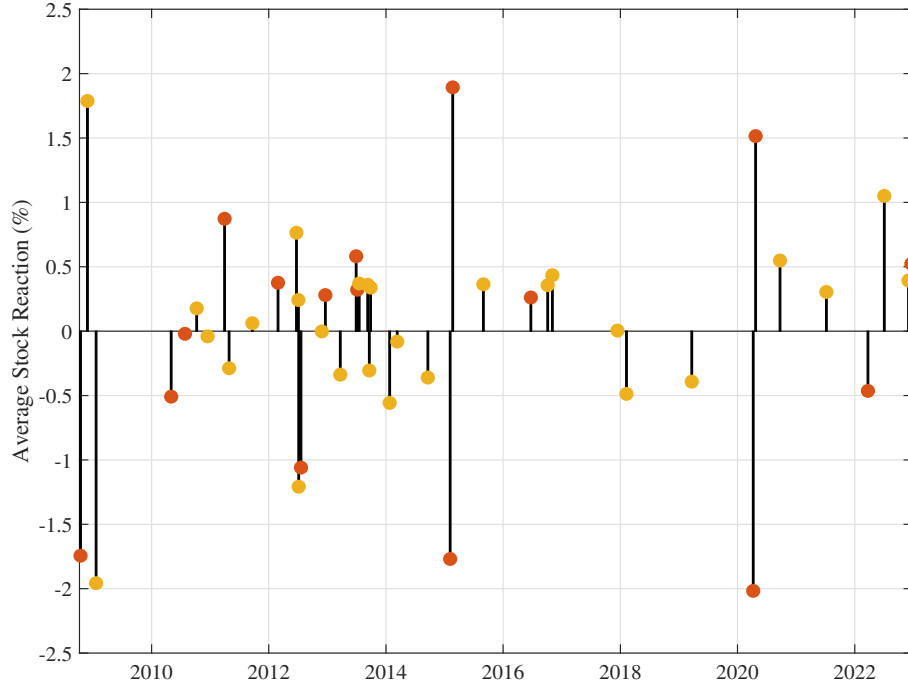


Figure 1: Collateral Policy Surprise Series: This figure plots the Collateral Policy Surprises, displayed in terms of the mean stock price reaction. Events coinciding with ECB governing council meetings and the 2012 euro summit are excluded, leaving a final sample of 43 events. Red dots indicate events affecting the collateral treatment of government bonds, while orange dots indicate events that exclusively affect the collateral treatment of assets issued by private agents.

3.3 Narrative Evidence

Our interpretation of collateral policy surprises as expansionary or contractionary relies on the direction of the bank stock response. The advantage of this approach is that we do not need to impose any interpretation on the direction of the surprise, but instead follow the assessment of market participants. To further corroborate the plausibility of this interpretation, we present a narrative assessment of selected episodes in the spirit of Drechsel and Miura (2026). We focus on a set of expansionary and contractionary events that are particularly relevant, as they are associated with average stock price reactions exceeding one percent in absolute value. These events correspond to the largest spikes observed in Figure 1.

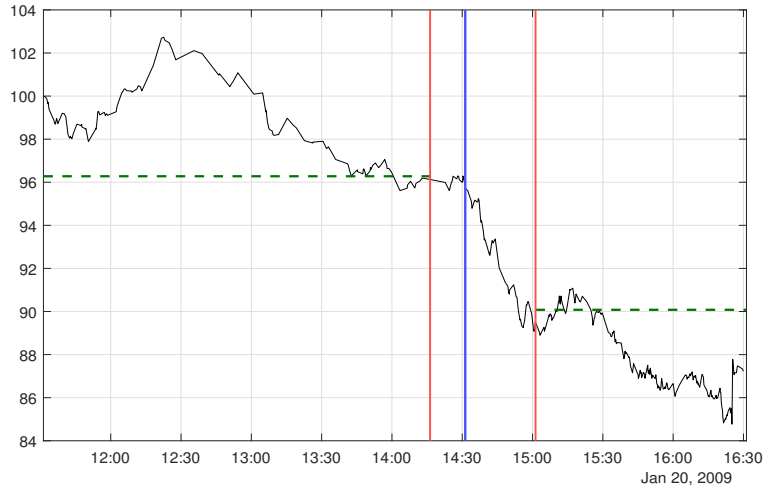


Figure 2: Negative high-frequency reaction: This figure plots the intraday first principal component of bank stock prices across all banks in our sample around a contractionary collateral policy surprise on Jan 20, 2009, normalized to 100 at the start of the window. The vertical blue line indicates the timing of the collateral policy announcement (press release). The vertical red lines mark the pre- and post-announcement windows. The horizontal dashed green lines indicate the median price levels within the pre- and post-announcement windows, expressed relative to the normalization at the start of the window. The horizontal axis reports time in UTC.

For example, on January 20, 2009, the ECB raised the minimum rating on asset-backed securities and uncovered bank bonds to AAA. This was accompanied by generally negative market commentary, stressing that it does not contribute to an improvement in market liquidity but rather results in greater financial stress. As Figure 2 shows, the aggregate bank stock return exhibits a sharp decline after the press release at 14:30, while there is very little movement in the 15 minutes prior to the event. The ECBs exclusion of Greek government bonds from the list of eligible collateral on July 20, 2012, offers another example: the announcement was accompanied by a strongly negative bank stock reaction and negative market commentary emphasizing the effect on bank funding cost. A similar pattern emerges on February 4, 2015. The ECB again removed Greek bonds from the list of eligible collateral, which had become eligible again in the meantime. The market commentary was again negative, mentioning concerns about runs on Greek government bonds.

Conversely, two important expansionary events concerning the treatment of Greek government bonds on February 20, 2015, and the Eurosystem’s response to the Covid-19 pandemic on April 22, 2020, are associated with a large positive bank stock reaction. Figure 3 shows the positive market reaction, the aggregate bank stock return exhibits a sharp jump after the press release in the late afternoon, and the stocks started trading again on a higher level the next day. In Appendix A.1, we provide a comprehensive assessment of the market commentary for each event, as well as the mean stock price reaction of the banks in our sample. This illustrates that the classification of collateral policy surprises into “expansionary” and “contractionary” events is generally consistent with the stock market response.



Figure 3: Positive high-frequency reaction: This figure plots the first principal component of bank stock prices across all banks in our sample around an expansionary collateral policy surprise on April 22, 2020, normalized to 100 at the start of the window. The vertical blue line indicates the timing of the collateral policy announcement (press release). The vertical red lines mark the pre- and post-announcement windows. The horizontal dashed green lines indicate the median price levels within the pre- and post-announcement windows, expressed relative to the normalization at the start of the window. The horizontal axis reports time in UTC.

- On January 20, 2009, the ECB says to require a rating of AAA/Aaa for ABS as collateral as of March 1st. Market commentary: *"From the banks' perspective, it's more restrictive, but it's not so drastic that it will shrink the mortgage market," said an analyst specialising in structured credit at a German bank. RBS economist Silvio Peruzzo said the announcement was minor compared to other changes announced to the ECB's collateral rules. "It provides an additional control on the quality of assets," he said. But Tullett Prebon economist Lena Komileva said the move sent the wrong signal at a time when the Bank of England and the U.S. Federal Reserve were stepping in to support illiquid markets. "Rather than responding to an underlying improvement in market liquidity conditions, the ECB's collateral tightening reflects quite the opposite – poor asset quality and insufficient market transparency – resulting in greater financial stresses," she said.* The bank stock reaction was unanimous and negative, and exceeded one percent in absolute terms.
- On July 20, 2012 the ECB announces that Greek government bonds become ineligible as collateral from July 25. Market commentary: *Greek bankers took the decision in their stride. "It's something we were expecting," one banker speaking on the condition of anonymity said. "The only difference is the borrowing cost for the banks."* The bank stock reaction was unanimous and negative, and exceeded one percent in absolute terms.
- On February 4, 2015, the ECB lifts waiver on credit rating requirements for Greek bonds. Market commentary: *ECB move rekindled safe-haven demand for low-risk U.S. and German government debt as it stoked worries about runs at Greek banks and concerns the new Greek government would have trouble renegotiating debt terms with its euro zone partners,*

analysts said. The bank stock reaction was unanimous and negative, and exceeded one percent in absolute terms.

- On February 20, 2015, the ECB is ready to reintroduce waiver for Greek collateral once it assesses that program likely to be concluded. The bank stock reaction was positive for every bank, and exceeded one percent in absolute terms.
- On April 22, 2020 the ECB says it takes steps to mitigate impact of possible rating downgrades on collateral availability. The bank stock reaction was unanimous and positive, and exceeded one percent in absolute terms.

3.4 Diagnostics

In this section, we present a large battery of plausibility checks for the collateral policy surprise series. We address concerns regarding the measurement of the surprise component, which are related to macro-financial confounders, predictability of the surprises, bank heterogeneity, and information effects.

Macro-Financial Confounders The measured CPS series would also be biased if bank stock price reactions captured not only the pure surprise component of collateral policy changes, but also the arrival of other information. To ensure that our event window is not contaminated by other major economic news, we check for overlapping data releases using the Bloomberg economic calendar, see Gürkaynak et al. (2020) and Altavilla et al. (2017). We find no significant macroeconomic announcements or monetary policy decisions in the United States, the euro area, or the United Kingdom in the same windows.

The measured CPS series would also be biased if it contains the systematic response of collateral policy to the macro-financial environment in addition to new information, i.e., if financial market participants fail to anticipate how collateral policy predictably responds to macroeconomic and financial conditions. Heider et al. (2015) and Allen et al. (2009) study such endogenous responses and the relationship with measures of interbank market stress through the lenses of theoretical models. There are two main arguments against such a bias. First, the Eurosystem’s economic trade-offs behind the collateral framework are publicly available and can be assumed to be common knowledge to market participants in general and to banks in particular. The ECB and other Eurosystem central banks frequently publish occasional papers rationalizing specific collateral policy changes. Bindseil et al. (2017) provide a summary of the economic trade-offs behind the ECB collateral policy and how it evolved over time. Second, the market commentary suggests that participants do track these systematic responses, implying that asset prices already reflect them and that the surprise component is isolated correctly. Nevertheless, it is instructive to correct the CPS for established measures of financial market stress.

Predictability of Collateral Policy Surprises Assessing whether the measured collateral policy surprise is predictable from pre-announcement information is important for its interpretation as an exogenous policy innovation. If the central bank systematically adjusts its collateral framework in response to sovereign-market fragmentation, financial stress, or incoming macroeconomic information, the resulting surprise may be correlated ex post with conditions observed before the announcement, even if market participants could not have predicted it in real time. This can arise if market participants do not fully understand the central banks’ policy reaction function (Bauer and Swanson, 2023). Our results, however, show that the collateral policy surprise series does not exhibit statistically significant predictability, even when it is regressed on a broad set of potential financial-market, macroeconomic, and monetary-policy predictors.

To examine whether observable financial and macroeconomic conditions predict collateral policy surprises, following Bauer and Swanson (2023), we estimate regressions of the following form:

$$cps_t = \alpha + \beta' \mathbf{X}_{t-}^{(h)} + u_t, \quad (7)$$

where t indexes the collateral policy announcement dates in our sample and cps_t denotes the collateral policy surprise associated with announcement t . The parameter α is a constant, while u_t is a regression residual. The vector $\mathbf{X}_{t-}^{(h)}$ contains information available immediately prior to collateral policy announcement t . For daily financial-market variables, the subscript $t-$ denotes the change over the preceding h trading days, calculated from day $t - h$ to day $t - 1$, where $t - 1$ is the last trading day before the announcement.⁷ For macroeconomic data surprises and monetary policy surprise factors, $t-$ denotes the most recently observed surprise value available just before the collateral policy announcement.

More specifically, $\mathbf{X}_{t-}^{(h)}$ includes changes in the first principal component of periphery-country 10-year sovereign yield spreads relative to Germany, changes in the German Bund yield-curve slope (defined as the 10-year yield minus the 1-year yield). It also contains percentage changes in the EUR/USD exchange rate, the EURO STOXX 50, and the Bloomberg Commodity Index, together with the change in the VSTOXX. The vector also includes the surprise components of the German ZEW Economic Sentiment and euro-area industrial production data releases, obtained from Bloomberg and measured as the difference between the released value and the corresponding market consensus expectation available immediately before the release. Collectively, these variables capture sovereign-market stress, broader financial conditions, and business-cycle developments.⁸

⁷A 65-trading-day window corresponds approximately to three calendar months and follows the baseline specification in Bauer and Swanson (2023). As a sensitivity check, we also report results using a shorter 10-trading-day window.

⁸These predictors were selected from a substantially larger set of financial and macroeconomic variables on the basis of bivariate regressions with the collateral policy surprise. We retained variables with an R^2 exceeding 0.25%. The initial set included changes and levels of financial and macroeconomic variables; the surprise components of euro-area, Germany, France, Italy and Spain macroeconomic data releases, including industrial production, GDP, unemployment, survey indicators, and consumer prices; and revisions to professional forecasts of euro-area macroeconomic and financial conditions obtained from Consensus Economics.

The vector $\mathbf{X}_{t-}^{(h)}$ additionally includes the most recent pre-announcement values of the conventional interest rate policy (Target), Timing, forward guidance, and quantitative easing surprise factors identified by Altavilla et al. (2019). These factors capture high-frequency surprises in the ECB’s current policy rate, the timing and expected path of future rate changes, and large-scale asset-purchase policies, respectively. Including these factors also allows us to test whether ECB monetary policy surprises predict our collateral policy surprises.

Table 1 reports estimates of equation (7). Columns (1) and (2) use 65-day changes in the daily financial variables, while columns (3) and (4) use 10-day changes. The baseline specifications in columns (1) and (3) include the financial-market and macroeconomic predictors, whereas columns (2) and (4) additionally control for the Target, Timing, Path, and LSAP monetary policy surprise factors. All regressions are estimated using 43 collateral policy announcements, and heteroskedasticity-robust standard errors are reported in parentheses.

Overall, the results provide little evidence that collateral policy surprises are systematically predictable using information available to financial markets before the announcement. Although a small number of individual coefficients are statistically significant, their significance is not generally robust across horizons and specifications, and the predictors are not jointly significant in any of the four regressions. The adjusted R^2 values are correspondingly low: the largest value, 0.09 in column (3), indicates that the model explains only about 10% of the variation in collateral policy surprises after accounting for the number of regressors. Given the small sample and the lack of joint statistical significance, this represents modest explanatory power rather than strong evidence of predictability.⁹ The results therefore support the interpretation that the collateral policy surprise series is largely orthogonal to observable pre-announcement financial, macroeconomic, and monetary policy conditions.

Information Effects As a next step, we address the potential role of central bank “information effects”, which have been suggested as a confounding factor in high-frequency monetary policy surprise measures (see Nakamura and Steinsson, 2018; Jarociński and Karadi, 2020; Miranda-Agrippino and Ricco, 2021). In our context, it is also important to highlight that ECB announcements are typically of a rather technical nature, as illustrated by the comprehensive list of collateral announcements provided in Appendix A.1. Nevertheless, market participants might uncover valuable information from such a policy announcement, for instance, about the quality of eligible assets or the role they play in the liquidity status of specific banks. Drechsel and Miura (2026) discuss information effects in the context of high-frequency bank stock prices to identify macroprudential policy shocks.

To shed light on this issue, we exploit the fact that our stylized model implies that a “pure” expansionary CPS should move both bank stocks and CDS-spreads in opposite directions, as it reduces the risk of bank failure due to liquidity shocks. In Figure 4, we compare the high-

⁹As a robustness check, we re-estimate our main results using a collateral policy surprise series orthogonalized with respect to the predictors in specification (3) in Table 1. The resulting estimates are very similar to the baseline results and are reported in Appendix B.3.

	<i>Dependent variable: collateral policy surprise</i>			
	65-day changes		10-day changes	
	(1)	(2)	(3)	(4)
Δ Periphery 10Y spread	−1.36 (0.99)	−1.43 (1.08)	−3.11 (2.29)	−3.95* (2.33)
Δ Bund yield curve slope	−0.14 (1.59)	0.06 (1.39)	−4.77 (3.93)	−6.08 (4.16)
Δ EUR/USD	−0.06 (0.15)	−0.02 (0.15)	0.28 (0.34)	0.28 (0.36)
Δ EURO STOXX 50	−0.09 (0.10)	−0.17** (0.08)	0.01 (0.22)	0.04 (0.22)
Δ VSTOXX	−0.02 (0.03)	−0.03 (0.02)	−0.03 (0.03)	−0.03 (0.04)
Δ BCOM	0.02 (0.06)	0.05 (0.05)	−0.15 (0.17)	−0.21 (0.19)
German ZEW surprise	−0.63 (0.59)	−0.52 (0.53)	−0.43 (0.51)	−0.36 (0.51)
Euro-area IP surprise	0.59* (0.33)	0.68** (0.32)	0.57** (0.24)	0.58* (0.32)
Target factor		−0.14 (0.22)		−0.06 (0.21)
Timing factor		−0.26 (0.35)		−0.10 (0.29)
Path factor		0.19* (0.11)		0.20* (0.12)
LSAP factor		−0.15 (0.33)		0.02 (0.37)
Adj. R^2	0.00	−0.04	0.09	0.04
Observations	43	43	43	43

Table 1: Predictability Regressions. Columns (1) and (2) use 65-day changes in the daily financial variables, whereas columns (3) and (4) use 10-day changes. The baseline specifications in columns (1) and (3) exclude the Target, Timing, Path, and LSAP variables; the full specifications in columns (2) and (4) include them. The symbol Δ denotes the change over horizon $(t - 1)$ to $(t - h)$ for the daily financial variables. Coefficient estimates are reported with heteroskedasticity-robust standard errors in parentheses. All specifications include a constant, which is omitted from the table for space considerations. The symbols *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

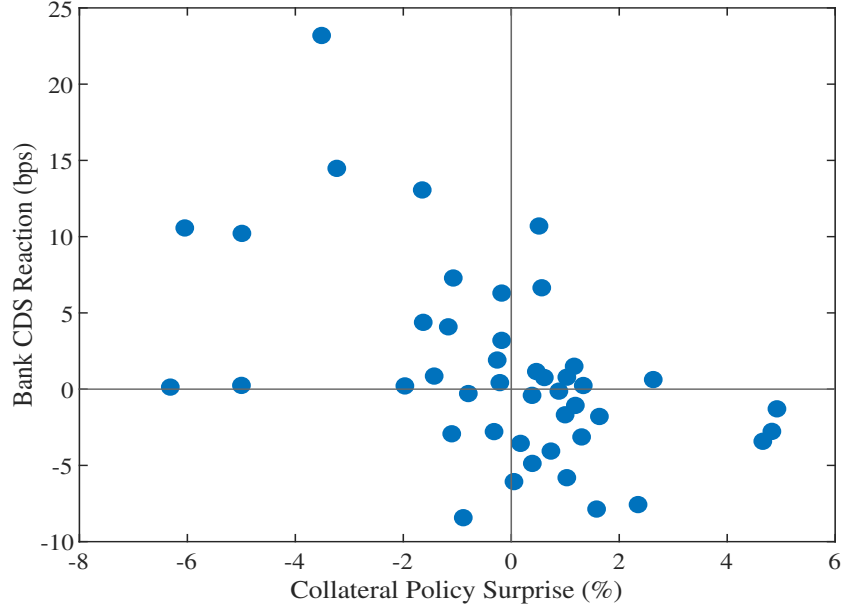


Figure 4: Information Effects: Bank Stocks and CDS Spreads. The figure plots the baseline collateral policy surprise based on the first principal component of high-frequency bank stock price reactions (x-axis) against the aggregate daily bank CDS reaction, measured in basis points.

frequency identified collateral policy surprises to the aggregate bank CDS reaction. On most events, bank stocks and CDS spreads indeed move in opposite directions. There are around 10 events in the southwest and northeast quadrant, but the stock price reactions are generally comparatively small, rarely exceeding one percent. Manually inspecting these events, it turns out that around half of them are of a very technical nature. For instance, five events concern information disclosure requirements on asset-backed securities or the introduction of a new collateral management system. Furthermore, there is hardly any market commentary on those event days, suggesting that information effects are not a concern. Excluding all events during which bank stocks and CDS spreads move in the same direction does not change the results.

Placebo Test A potential concern regarding the high-frequency approach can be that other news might affect the bank stock price during the event window, even if it is intraday. To test this, we conduct a “placebo” test by measuring intraday changes in bank stock prices using the same event window, but shifted to 7 days before each collateral policy announcement. These days serve as a proxy for comparable days without any actual collateral policy announcements. We do not find any significant changes in financial variables in response to these “placebo” surprise series. This can be illustrated graphically via probability density functions similar to Känzig (2021). Figure 5 plots the kernel density estimate for our baseline surprise series against the kernel density estimate for the “placebo” surprise series. The bank stock return distribution on event days (solid black line) shows increased variance and fatter tails on announcement event windows as opposed to control dates (dashed black line).

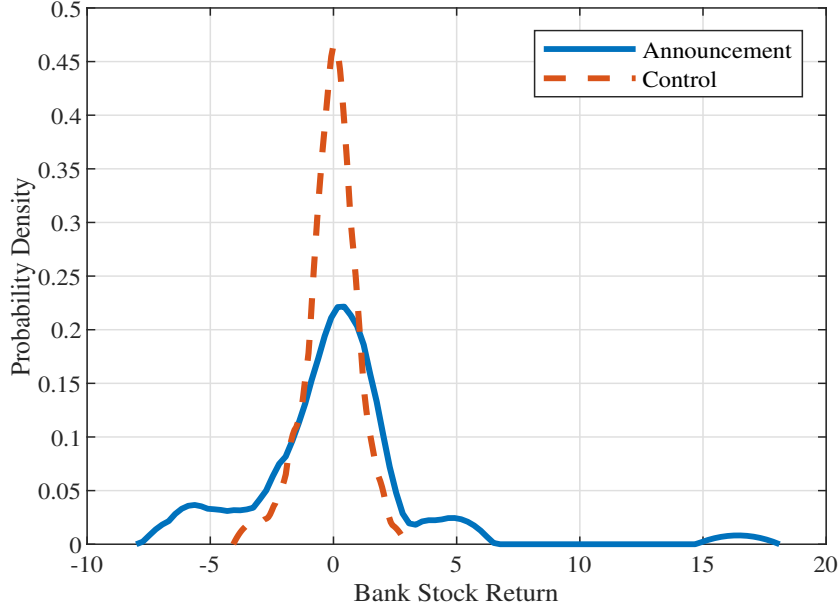


Figure 5: Announcement versus Control Event Window. The figure plots the Epanechnikov kernel density estimate of aggregate bank stock returns on our 43 event dates (solid black line). As a control group, we report the Epanechnikov kernel density estimate for a placebo surprise series that is based on the same intraday 45-minute event window, but 7 trading days before each actual collateral announcement.

3.5 Bank Heterogeneity and Measurement

While our measured collateral policy surprise series has appealing properties from a macro-econometric point of view, there might be concerns about bank selection and heterogeneity as potential confounders of our measurement of collateral policy surprises. As a first step, we demonstrate that our measure of collateral policy surprises, based on high-frequency stock price data, is robust to the selection of banks. In Appendix A.3, we show that excluding any one bank at a time does not affect our shock series. We also illustrate that for most events with a positive (negative) first principal component, the maximum and minimum stock price reactions were also positive (negative). By contrast, the dispersion in bank stock reactions is not visibly correlated with the aggregate response. This implies that purging the CPS by the bank return dispersion has no relevant effect on the measurement of collateral policy.

Bank Balance Sheet Characteristics To explore cross-sectional heterogeneity in a more structural fashion, we obtain balance sheet data from CapitalIQ for all banks included in our intraday sample, spanning the period from Q1 2007 until Q4 2024. We estimate the following regression at the bank-quarter level:

$$\Delta \text{Stock}_{i,t} = \alpha + \beta \text{weak}_{i,t-1} + \gamma \mathbf{X}_{i,t} + \phi_i + \tau_t + \epsilon_{i,t} \quad (8)$$

where ΔStock is the (high-frequency) stock price reaction around a collateral policy change of bank i , cumulated over all events in quarter t . We define “weak” in terms of liquidity and solvency, respectively. First, we define a bank to be weak in terms of liquidity if its securities-

over-assets-ratio is below the median of all the banks in the sample. Second, we define a bank to be weakly capitalized if its tier-one equity ratio is below the median of all banks in the sample. We expect that $\beta > 0$ as collateral policy changes are stronger for banks with less liquid / less capitalized banks, following Kashyap and Stein (2000). We also include a vector of time-varying control variables at the bank level $\mathbf{X}_{i,t}$. Here, $\ln(\text{Assets})$ is the log of total assets; Loans Ratio is the loans-over-assets ratio; ROE is the return on equity; Debt Ratio is the total debt-over-assets ratio; Deposits Ratio is the deposits-over-assets ratio of bank i in quarter t . Summary statistics can be found in Table A.3 in Appendix A.3. We further include bank ϕ_i and quarter τ_t fixed effects.

	(1)	(2)	(3)	(4)	(5)	(6)
	$\Delta \text{ Stock}_i$	$\Delta \text{ Stock}_i$	$\Delta \text{ Stock}_i$	$\Delta \text{ Stock}_i$	$\Delta \text{ Stock}_i$	$\Delta \text{ Stock}_i$
Illiquid $(0/1)_{i,t-1}$	0.286 (0.240)	0.237 (0.322)	0.396** (0.176)			
Low Capital $(0/1)_{i,t-1}$				0.328 (0.220)	0.304 (0.406)	0.192* (0.095)
$\ln(\text{Assets})$	-0.009 (0.062)	-0.131 (0.447)	-0.921* (0.496)	-0.047 (0.080)	-0.287 (0.436)	-0.865* (0.411)
Loans Ratio	0.006 (0.013)	-0.021 (0.023)	-0.016 (0.011)	0.007 (0.012)	-0.023 (0.025)	-0.015 (0.009)
ROE	-0.038 (0.028)	-0.039 (0.031)	-0.021 (0.023)	-0.036 (0.028)	-0.036 (0.031)	-0.019 (0.023)
Debt Ratio	0.014 (0.011)	0.066** (0.023)	0.003 (0.006)	0.017 (0.010)	0.066** (0.024)	0.003 (0.007)
Deposits Ratio	-0.005 (0.012)	0.033* (0.016)	-0.008 (0.012)	-0.005 (0.011)	0.029 (0.018)	-0.008 (0.012)
Observations	231	231	231	231	231	231
R-squared	0.035	0.085	0.824	0.040	0.088	0.820
Bank FE	No	Yes	Yes	No	Yes	Yes
Quarter FE	No	No	Yes	No	No	Yes
Cluster	Bank	Bank	Bank	Bank	Bank	Bank

Table 2: Collateral Policy Surprises and Bank Heterogeneity. This table presents the results from estimating equation (8) over all collateral policy events. ***, **, * indicate significance at the 1, 5, and 10% level.

Table 2 presents the results from estimating (8) over all collateral policy events. Columns (1)-(3) show that less liquid banks react more to collateral policy surprises. Importantly, the variation in individual stock price reactions seems to be driven by the common aggregate response to each event, which is reflected by an R^2 of more than 0.8 once quarter fixed effects are included. By contrast, the R^2 is less than 0.1 in specifications without quarter fixed effects. Furthermore,

significant cross-sectional differences emerge only after partialling out the common component. Similarly, columns (4)-(6) show that less capitalized banks also seem to react more to collateral policy changes, albeit with a lower magnitude. Since collateral policy is primarily associated with banks' short-term funding needs, it is reasonable to expect that bank liquidity is a more important characteristic than bank capitalization. Overall, these results are in line with Kashyap and Stein (2000) who find similar results for changes in monetary policy. Collateral policy surprises predominantly affect aggregate liquidity, and the cross-sectional variation appears to be of second-order importance for the *measurement* of collateral policy surprises.

4 The Transmission of Collateral Policy Surprises

In this section, we use our high-frequency collateral policy surprise (CPS) series to study the transmission of collateral policy to general financial markets, bank CDS spreads, and the euro area government bond market. Throughout the analysis, we estimate the dynamic responses of asset prices to collateral policy surprises at daily frequency.

4.1 Financial Markets

We estimate the effect on key financial market variables separately:

$$y_{t+h} - y_{t-1} = \alpha + \beta_h cps_t + \epsilon_t, \quad (9)$$

where t indexes collateral policy announcements, h is the number of days after the announcement and ranges from $h = -1$ to $h = 10$, cps_t is the high-frequency collateral policy surprise series measured using intraday data, y_{t+h} is the asset price h days after the announcement, y_{t-1} is the asset price one day before the announcement, and ϵ_t is the residual. We estimate both the immediate and the delayed effects of policy surprises on asset prices, examining (i) the contemporaneous asset price response on the announcement day and (ii) the lagged asset price responses over subsequent trading days. As customary in the local projections literature, each time horizon is estimated separately. Standard errors are adjusted for serial correlation and heteroskedasticity.

Figure 6 presents the effects of an expansionary one-standard-deviation collateral policy surprise (CPS) on key financial market indicators. Following an expansionary CPS, the EURO STOXX 50 index increases significantly, indicating a broad-based positive market reaction. As expected, the EURO STOXX Banks index also reacts positively. Crucially, its response is markedly stronger than that of the overall stock market, as shown by Bank Excess Returns, the middle panel, of Figure 6. To put the numbers into perspective, a one-standard-deviation monetary policy surprise (Jarociński and Karadi, 2020) raises stock market returns by 0.4 percentage points, with no differential effect on bank stocks relative to the overall stock market; see also Table 5 below. We also find that expansionary collateral policy significantly reduces the implied volatility of EuroSTOXX50 options. This measure, known as V2X, is a common

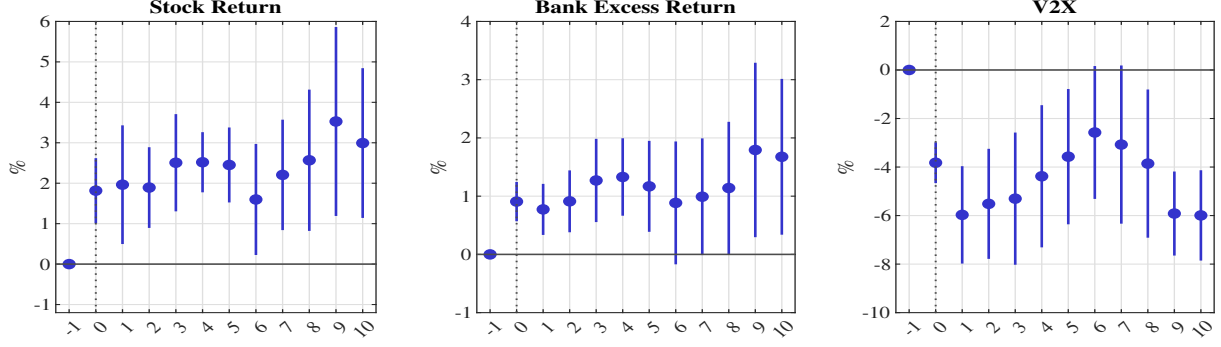


Figure 6: Key Financial Indicators: Results from estimating equation (9) using all collateral policy surprises. The dependent variables are the EuroSTOXX50 return, the excess return of EuroSTOXX Bank Index over the overall index, and the implied volatility of EuroSTOXX50 options (V2X). Dots represent estimated coefficients, and vertical lines denote 90% confidence intervals based on heteroskedasticity- and autocorrelation-consistent (Newey-West) standard errors. The horizontal axis reports event time in days relative to the policy announcements, with $t = 0$ marking the announcement dates.

proxy for European equity market uncertainty. The magnitude of both effects remains stable or even strengthens over time, suggesting a persistent dampening of financial market uncertainty. Furthermore, our results are not confined to data at daily frequency. In Appendix B.5, we aggregate the data up to a monthly frequency and find quantitatively similar results. Lastly, in Appendix B.3, we show that the asset prices used as outcome variables exhibit no significant pre-trends prior to the collateral policy announcements.

4.2 Government Bond Market

We now turn to the government bond market effects of an expansionary CPS. Since there is no broadly accepted safe asset in the euro area, we examine different euro area government bonds and uncover substantial heterogeneity in the transmission. There are several possible explanations for such heterogeneity. For example, collateral policy might heterogeneously affect the convenience yield of government bonds. By raising their collateral value, an expansionary CPS directly raises the price of affected bonds and lowers safe scarcity in general equilibrium. Since sovereign CDS do not contain a collateral premium, an expansionary CPS lowers the CDS-bond basis. Alternatively, intermediary asset pricing frameworks predict that an expansionary CPS lowers banks' default risk, see Appendix A.2 for details. Lower default risk makes banks effectively discount the future by less and, thereby, raises their willingness to pay for any asset, including protection against government default, i.e., sovereign CDS. This implies a uniform reaction in bond and CDS spreads, without affecting the CDS-bond basis.

As a first step, we examine the responses of 5-year government bond spreads over the risk-free rate, proxied by the Overnight Index Swap (OIS). We focus on the eight largest euro area sovereign borrowers, which have reliable bond market data at daily frequencies. Following the literature on the financial market effects of monetary policy (Swanson, 2021; Bauer et al., 2023),

we estimate local projections

$$y_{t+h,i} - y_{t-1,i} = \alpha + \beta_h^c cps_t D_i + \beta_h^p cps_t (1 - D_i) + \epsilon_{t,i} , \quad (10)$$

where the dummy variable D_i equals one for countries located in the core and zero for countries located in the Periphery. All other variables are defined as in Equation (9). In the main text, we focus on the results when countries are split into these two groups. The responses of each individual country are presented in Appendix B.

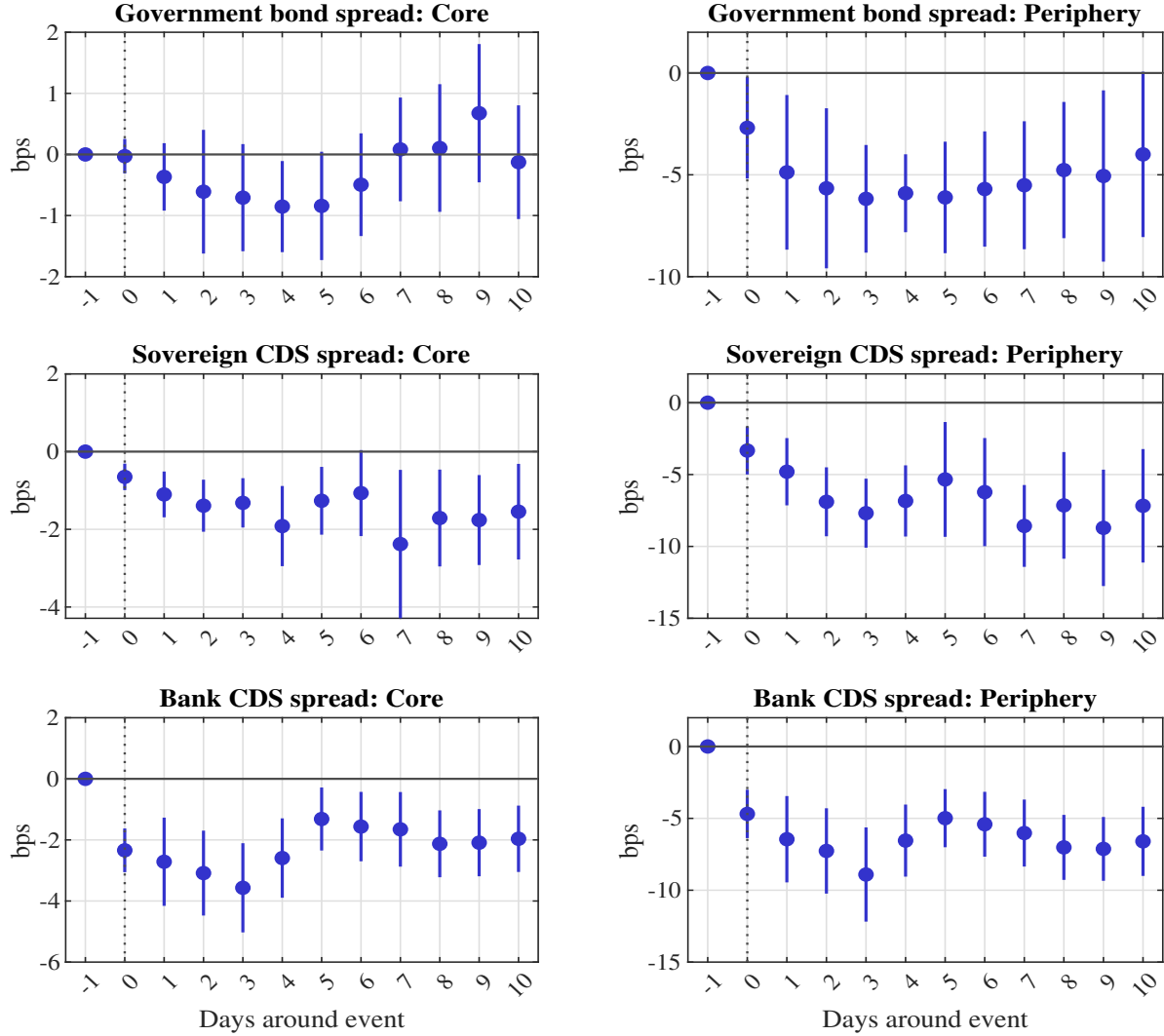


Figure 7: Transmission to Government Bond Market and Bank Default Risk: Results from estimating equation (10) on the full sample. The top row reports 5-year government bond yield spreads relative to maturity-matched OIS. The middle row reports 5-year sovereign CDS spreads. Core countries are Germany and France; periphery countries are Italy and Spain. The bottom row reports 5-year bank CDS spreads, split based on bank location. Dots represent estimated coefficients and vertical lines denote 90% confidence intervals based on heteroskedasticity- and autocorrelation-consistent (Newey-West) standard errors. The horizontal axis reports event time in days relative to the policy announcements, with $t = 0$ marking the announcement dates.

The top row of Figure 7 shows the results. While the effects on core bond spreads are marginally significant and economically modest at almost 2 basis points, the bond spread in

periphery countries declines by around 8 basis points. Consequently, the periphery-core spread declines significantly after an expansionary CPS for every estimation horizon. The effect is quite sizable and peaks at around -6 basis points several trading days after the shock. We find no significant effects on OIS spreads themselves, suggesting that the sovereign bond market responses primarily reflect changes in periphery bond yields, in both magnitude and statistical significance. Similar to the financial market results, we show in Appendix B.5 that the government bond market is also affected at monthly frequencies. For instance, the periphery bond and sovereign CDS spreads drop by around 15 basis points, while core spreads do not move significantly.

Next, we examine whether these heterogeneous effects on government bond spreads carry over to the pricing of sovereign risk more generally. Specifically, we obtain five-year CDS spreads for the largest four euro area countries (Germany, France, Italy, and Spain), which represent the most commonly traded CDS tenor. In the middle row of Figure 7, we observe a significantly negative effect of expansionary collateral policy on sovereign CDS spreads. Again, the effects are much more pronounced for riskier countries (Spain and Italy), while they amount to slightly more than one basis point for core countries (France and Germany). The responses of individual countries are shown in Figure B.2 in Appendix B.1.

The decline in sovereign CDS is quantitatively very similar to the decline in government bond spreads. Consistent with this observation, we show in Appendix B that the CDS-bond basis does not respond significantly to collateral policy surprises. Taken together with the positive effects of expansionary collateral policy on aggregate uncertainty, these results are at odds with the traditional convenience yield channel, which would instead predict a basis response concentrated in bond markets. Instead, our results point towards a bank default risk channel of collateral policy. Expansionary collateral policy surprises raise banks' willingness to pay for any asset, including sovereign bonds and sovereign CDS. This drives down government bond and CDS spreads for the same country uniformly.

4.3 Bank CDS

The evidence presented so far is consistent with a transmission channel from collateral policy to government bond markets via the banking sector. To further unpack the heterogeneous transmission of collateral policy to government bonds and sovereign CDS, we examine banks' CDS responses in the bottom row of Figure 7. We subset banks into "Core" and "Periphery" based on their headquarters' location. For completeness, we report the results on individual bank CDS spreads in Figure B.3 in Appendix B. While the effects of expansionary collateral policy is significantly negative for all banks, it is much more pronounced for banks located in the periphery. Periphery banks react considerably more strongly, and these banks were also holding more periphery sovereign bonds during our sample period.

As Table 3 illustrates, core and periphery banks differ along several dimensions. Periphery banks have higher CDS spreads, a smaller share of liquid assets, less tier 1 capital, and a larger

home bias in their government bond holdings. Information on European banks’ home bias can be obtained from publicly available data from the European Banking Authority’s transparency exercise in 2013.¹⁰ Consequently, a similar picture emerges when splitting banks at the median for each of these dimensions. Our analysis suggests that the source of the uneven transmission of collateral policy lies in bank heterogeneity. Indeed, an intermediary asset pricing framework that also accounts for banks’ home bias in sovereign bond holdings predicts an uneven transmission of a symmetric collateral policy surprise to the government bond market.

Variable	Full Sample	Core	Periphery
CDS Spread (bps)	177.11	136.00	228.49
Securities Ratio (%)	41.76	53.18	27.49
Tier-1 Ratio (%)	9.30	9.73	8.75
Home Bias (%)	63	51	78

Table 3: Euro Area Bank Heterogeneity. This table presents summary statistics for core and periphery banks. The CDS-spread refers to the full-sample average from 2008-2022. Home bias in sovereign bond holdings is obtained from the 2013 EBA stress test. The securities and tier-1 capital ratios are computed with respect to total assets, also computed in 2013.

4.4 Sample Splits

In this section, we examine how the transmission of collateral policy differs across specific sub-samples. We begin by differentiating between events that affect either the collateral treatment of government bonds or of privately issued debt. This sample split is informative about the starkly different reactions of core and periphery government bond yields. Building on the evidence with respect to bank heterogeneity, we also study events directly related to the sovereign-bank nexus, before discussing the role of the business cycle and central bank asset purchases.

Non-Sovereign vs Sovereign Events Almost half of all collateral policy changes refer to government bonds, which play a central role in banks’ collateral management due to their extraordinary safety and liquidity. To investigate whether the transmission differs across these two groups, Column (1) in Table 4 focuses on events that only affect private sector assets, leaving the collateral treatment of government bonds unaffected. Column (2), by contrast, considers events that also or exclusively affect government bonds.

We find that financial market responses are generally similar across sub-samples. However, when turning to the government bond market, periphery spreads decline by 4 basis points, compared to half in the sample related to non-sovereign events. The reaction of core government bond spreads is insignificant across both sub-samples. Similar effect sizes can be observed for sovereign CDS spreads: periphery sovereign CDS declines by more than 5 basis points for sovereign events, compared to 2 basis points for non-sovereign events. For the core, sovereign

¹⁰Details on the EBA’s 2013 EU-wide transparency exercise, and more recently, the 2025 exercise, can be found under [this link](#).

CDS decline by 1 basis point, compared to almost flat in the non-sovereign sample. In the bottom rows, we also find the largest declines in bank CDS for banks headquartered in the periphery, irrespective of the type of events.

Sovereign-Bank Nexus Events The heterogeneous transmission of collateral policy to the government bond market is particularly pronounced for events directly related to the sovereign-bank nexus. Out of the 43 events, 15 events affect government bonds directly. These events are indicated with a "1" in the last column of Table A.1. The remaining 28 events are non-sovereign events and are related to changes in privately issued assets. These events are marked with a "0". Furthermore, we split events into those with a clear connection to the sovereign-bank nexus and indicate them by an asterisk in the last column of Table A.1. Some of the sovereign-bond events are marked with an asterisk, as they directly targeted sovereign borrowers most affected by the euro area sovereign debt crisis. Several of the remaining non-sovereign events also clearly relate to the sovereign-bank nexus, for instance, through changes to collateral eligibility for unsecured or government-guaranteed bank debt and asset-backed securities that made up a large share of euro area periphery banks' portfolios.

To provide intuition for the nature of collateral policy changes related to the sovereign-bank nexus, we consider three individual events that triggered large movements in bank stocks, exceeding one percent over the 45-minute event window. On 31 March 2011, the ECB suspended the minimum rating requirement on Irish government bonds, making them eligible as collateral. This announcement was associated with an increase in aggregate bank stock prices and a narrowing of the 5-year Italian–German government bond spread by 4.2 basis points within one trading day. By contrast, on 20 July 2012, the ECB excluded Greek government bonds from the list of eligible collateral, inducing a decline in bank stocks and a widening of the Italian–German spread by 37.5 basis points within a single day. This contractionary collateral policy change occurred during a period when periphery bond spreads exceeded 500 basis points. Finally, on 20 February 2015, the ECB reinstated Greek government bonds as collateral, which was associated with positive bank stock returns and a reduction of the Italian–German spread by 6.4 basis points.

These events are noteworthy because they affect only the collateral treatment of risky sovereign debt while leaving haircuts on safe bonds, such as the German bund, unchanged. As a result, the collateral value of periphery government bonds increases relative to core bonds, which in turn transmits heterogeneously to government bond prices. At the same time, such a policy is particularly beneficial for periphery banks, which hold a large share of domestic sovereign bonds. Their ability to borrow from the central bank drastically improves after the shock, leading to pronounced declines in bank CDS spreads.

Additionally, some events clearly affect banks in euro area periphery countries that were closely associated with the sovereign-bank nexus at that time. For instance, on 21 September 2011, the Eurosystem reduced the limit for using unsecured debt instruments issued by banks as collateral, which was particularly relevant for periphery banks. On 22 June 2012, the ECB

Variable	Full Sample		Government Bonds		Sovereign-Bank Nexus		Business Cycle		Central Bank APP	
	(1)	Unrel. (2)	Related (3)	Unrel. (4)	Related (5)	Boom (6)	Recession (7)	Pre (8)	Post (9)	
Stock Return	1.8***	1.8***	2.0***	1.7***	2.1***	0.6***	2.8***	2.2***	1.0	
Bank Excess Return	0.9***	1.1***	0.9**	0.7**	1.3***	0.3**	1.5***	1.0***	0.7*	
V2X	-3.8***	-3.8**	-4.5**	-3.4**	-4.5***	-2.2	-4.7**	-4.3***	-3.6	
Sov Spread (Core)	0	0.4	-0.7	0.7	-1.1*	0.3**	-0.4	-0.0	-0.2	
Sov Spread (Periphery)	-2.7***	-1.8*	-4.1**	-1.0	-5.2**	-0.6	-4.0**	-3.1**	-1.7***	
Sov CDS (Core)	-0.7***	-0.3*	-1.1***	-0.4**	-1.0***	-0.3**	-1.0***	-0.8***	-0.4	
Sov CDS (Periphery)	-3.3***	-2.1**	-5.3***	-2.3**	-4.9***	-1.8***	-4.4***	-4.2***	-1.3*	
Bank CDS (Core)	-2.3***	-2.5***	-2.4**	-2.0***	-2.9***	-1.0***	-3.2***	-3.1***	-0.6***	
Bank CDS (Periphery)	-4.7***	-5.1***	-5.0***	-4.0***	-5.8***	-1.6***	-6.8***	-6.2***	-1.4**	
Number of events	43	28	15	26	17	26	17	29	14	

Table 4: Collateral Policy Surprises in Different Sample Splits. The table reports the contemporaneous response of key variables to a one standard deviation expansionary CPS for different sample splits. Column (1) reports the full sample results. Columns (2) and (3) report the results for events affecting government bonds and other assets. Columns (4) and (5) report results related to the sovereign-bank nexus; columns (6) and (7) report results for booms and recessions classified by the CEPR. Columns (8) and (9) report the results for Pre- and Post-APP events, with 9 March 2015 as the cut-off date. All yields and spreads are expressed in basis points. ***, **, * indicate significance at the 1, 5, and 10% level.

reduced the rating threshold for asset-backed securities. Market commentary on this day claims that this policy change was "designed to ease pressure on Spain". Furthermore, an anonymous trader associates this measure with narrowing core-periphery government bond spreads. Similarly, on 18 July 2013, the ECB started to accept a broader set of asset-backed securities. The market commentary on this day explicitly mentions two Italian banks that exhibited large stock price increases, noting that French, Italian, and Spanish banks were particularly affected by this policy change.

Columns (4) and (5) in Table 4 present the results for the sovereign-bank nexus split. Stock market reactions are again similar across the sample splits. In the bond market, periphery government bond spreads react most and decline by up to 5 basis points for related events, while core government bond spreads react by merely up to 1 basis point. Similarly, periphery sovereign CDS declines by almost 5 basis points for related events, while core sovereign CDS declines by 1 basis point. As before, with almost 6 basis points, the largest bank CDS declines occur for periphery banks.

This suggests that the sovereign-bank nexus, rather than the targeted asset class itself, is the more relevant dimension along which collateral policy transmission varies, irrespective of the economic context. While we cannot attribute the effects of expansionary collateral policy to different components of the doom loop, such as bailout costs and their effect on sovereign risk or portfolio losses and their effect on bank risk, our results provide causal evidence that expansionary central bank collateral policy has strong positive effects on the pricing of sovereign and bank default risk, in particular during times of high stress.

Business Cycle Next, we investigate the role of the macroeconomic environment for the transmission of collateral policy. If adverse economic conditions are associated with elevated stress in the banking system, an expansionary CPS might have more pronounced effects on banks and the government bond market. For example, Brunetti et al. (2011) study the effects of central bank policy interventions on interbank market stress and find different effects in crisis and normal times. Hence, we separately estimate our baseline specification during periods classified as expansions and recessions by the CEPR.¹¹ In this context, it is important to recall that collateral policy typically accommodates the temporarily high demand for central bank funding during crisis times, which otherwise might be constrained by the limited availability of eligible collateral. However, our CPS isolates the unexpected change in collateral policy, so our results do not reflect the systematic responses of collateral policy to the macro-financial environment.

In Table 4, we summarize the contemporaneous effects of a collateral policy surprise. Consistent with a temporarily high liquidity demand during downturns, we observe that bank stocks react more strongly. The bank excess return is 1.5 percent during recessions, while it amounts to merely 0.3 percent during expansions. At the same time, collateral policy induces roughly double the decline in the V2X during recessions. Furthermore, bank CDS spreads decline by almost 7 basis points compared to merely 2 basis points during expansions. In line with the

¹¹The CEPR recession indicator can be found under [this link](#).

larger financial market effects, the heterogeneous transmission to core and periphery bonds is also more pronounced (around 3 basis points) than in expansions, where it is at most one basis point. Again, these responses are primarily driven by periphery yields. Similarly, periphery sovereign CDS spreads decline much more during recessions. Taken together, this suggests that the macroeconomic environment plays an important role in the transmission of collateral policy.

Asset Purchases and Excess Liquidity In addition to the macroeconomic environment and hence liquidity demand, it is also natural to examine the role of liquidity supply for the transmission of collateral policy. The overall amount of central bank liquidity in the form of excess reserves may soften the effects of collateral policy by reducing banks’ reliance on central bank facilities. Large-scale asset purchases started in March 2015 by the Eurosystem expand central bank reserves and reduce the demand for collateral. Consistent with this, we find stronger and highly significant effects for most variables in the period preceding these purchases (Column 5). Nevertheless, we still observe significant effects in the Post-APP sub-sample (Column 6), particularly for government bond yields and CDS spreads. This observation suggests that some banks continue to benefit from access to central bank facilities even in periods of elevated aggregate liquidity.

4.5 Comparison to Monetary Policy Surprises

To benchmark the effects of our newly identified collateral policy surprises against conventional monetary policy surprises, we estimate equation (10), using the monetary policy surprise series from Jarociński and Karadi (2020). An expansionary one standard deviation monetary policy surprise raises the STOXX 50 index by 0.4 percent and lowers nominal bond yields by around 2 basis points. We normalize the MPS to match the same STOXX 50 response as an expansionary one standard deviation CPS.

To further disentangle which dimension of conventional monetary policy the CPS is (or is not) comparable to, we additionally consider the Target, Timing, Forward Guidance, and QE factors of Altavilla et al. (2019), applying the “poor man’s” information-effect correction of Jarociński and Karadi (2020) to each factor separately, following Miranda-Agrippino and Nenova (2022). Table 5 presents the contemporaneous response (β_0): the CPS in Column (1), the composite MPS in Column (2), and the four monetary policy factors in Columns (3) to (6). For each factor, we report the response to an expansionary surprise. Due to their small and less significant effects on the stock market, the multi-dimensional factors are not normalized to the STOXX 50 reaction but reported in their own one standard deviation units. The results are robust across different estimation horizons.

Panel A of Table 5 reports financial market effects. Conventional monetary policy surprises have little effect on equity markets, irrespective of using the composite MPS or its individual Target, Timing, Forward Guidance, and QE dimensions. The EUROSTOXX 50 responds only modestly, significantly so for the composite MPS and the Forward Guidance and QE factors,

Variable	CPS (1)	MPS (2)	Multi-dimensional MP			
			Target (3)	Timing (4)	FG (5)	QE (6)
<i>Panel A: Financial Markets</i>						
STOXX 50	0.9***	0.9**	0.7	0.0	0.3**	0.2**
STOXX Banks	1.8***	0.9*	0.9	0.0	0.3	0.1
V2X	-3.8***	-2.7**	-1.4	-0.3	-0.5	-0.8
Bank CDS Spread	-2.3***	-0.5	-0.6	0.1	-0.3	-0.4
Bank CDS (Periphery)	-4.7***	-0.9	1.0	0.4*	-0.5	-0.5
<i>Panel B: Government Bond Market</i>						
Core Yield, 5Y	0.7**	-4.7***	-1.1**	-2.5***	-3.1***	-2.1***
Periphery Yield, 5Y	-1.6**	-3.8***	-2.2*	-2.5***	-2.5***	-1.8***
Periphery-Core Spread, 5Y	-2.3**	0.9***	-1.2	0.1	0.6	0.3
Germany CDS, 5Y	-0.4**	0.9	0.2	0.0	0.1	0.3
France CDS, 5Y	-0.9***	-0.7	-0.3	0.3*	0.3	0.4*
Italy CDS, 5Y	-3.6***	1.1	-2.2	0.1	0.2	0.8
Spain CDS, 5Y	-3.1***	3.4	-0.4	-0.3	1.2	0.9
<i>Panel C: International Transmission</i>						
EURUSD	0.2	-0.2**	-0.1	-0.2***	-0.3***	-0.3***

Table 5: Collateral and Monetary Policy Surprises. The table reports the contemporaneous response (β_0) to a one standard deviation expansionary collateral policy surprise (CPS), an expansionary monetary policy surprise (MPS, Jarociński and Karadi, 2020), and the expansionary Target, Timing, Forward Guidance (FG), and QE factors of Altavilla et al. (2019). The MPS is normalized to the same STOXX 50 reaction as the CPS; each dimension of monetary policy is scaled by its own standard deviation. *Panel A* shows results for financial markets; *Panel B* for the government bond market; *Panel C* for international transmission. All yields, spreads and CDS are in basis points. ***, **, * indicate significance at the 1, 5, and 10% levels.

while the EURO STOXX bank index shows no visible excess return over the wider market for any conventional measure. This is in sharp contrast to the expansionary CPS, which generates a pronounced bank excess return. While the composite MPS also lowers the V2X, the effect is smaller than that of the CPS, and none of the four individual factors moves the V2X significantly. Similarly, neither the composite MPS nor any of its dimensions affects bank CDS spreads. The one robust effect of conventional monetary policy is the well-documented impact on the exchange rate, present also for the composite MPS and the Timing, Forward Guidance, and QE factors, which is entirely absent for the collateral policy surprise. In Appendix B.4, we demonstrate that collateral policy surprises also do not materially affect CIP deviations and cross-border funding conditions more generally.

Turning to the government bond market in Panel B of Table 5, expansionary conventional monetary policy lowers government bond yields across the board: the composite MPS and all four factors significantly reduce both core and periphery yields. Crucially, however, the periphery-core spread does not compress: it is significantly positive for the composite MPS and statistically insignificant for each of the four dimensions. This is again qualitatively different from the heterogeneous, spread-compressing effect of the expansionary CPS in column (1). Consistent with this, we find no compression of sovereign CDS spreads under any conventional measure,

whereas the CPS lowers them markedly, especially for periphery sovereigns. Taken together, these results highlight that collateral policy operates through very different transmission channels than monetary policy – including each of its conventional dimensions – both across financial and government bond markets.

5 Conclusion

In this paper, we develop a high-frequency approach to identify the effects of central bank collateral policy on banks and the government bond market. We identify collateral policy surprises using bank stock price changes over short time windows around Eurosystem collateral announcements. By broadening access to central bank funding, expansionary collateral policies increase bank stock excess returns and reduce financial uncertainty indicators. We then show that expansionary collateral policy reduces government bond yields, in particular for the euro area periphery. A similarly uneven decline is observed for sovereign and bank CDS spreads, in line with the strong home bias of euro area banks. All effects are more pronounced for events affecting the sovereign-bank nexus, during times of economic stress, and in the pre-APP period, when liquidity was less abundant and central bank borrowing played a greater role. Our results suggest that the transmission of collateral policy differs across countries, and that this variation seems to be closely linked to fragmentation in the banking sector. These asymmetric effects on banks and government bond markets differ fundamentally from the effects of conventional monetary policy, which affect euro area countries uniformly.

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A Measurement of Collateral Policy Surprises

This section provides detailed information on the collateral policy events that received media coverage and are potential candidates to be included in our collateral policy surprise series. For each event, we know the exact time stamp of the ECB’s policy announcement. We summarize briefly the main content of each announcement and report relevant market commentary and additionally relevant background information in the third column of Table [A.1](#).

A.1 List of Collateral Policy Announcements

Table A.1: The table lists the *Date* and *Title* of ECB press releases concerning collateral policy announcements, together with corresponding market commentary. *Stock* reports the sign of the average intraday stock price change. *Sovereign* indicates whether the announcement is related to sovereign bonds. An asterisk (*) denotes events that affect the collateral treatment of an individual sovereign issuer, or the closely related treatment of bank debt that receives government guarantees. *Source:* LSEG Refinitiv and ECB website.

Date	Title	Market Commentary/Background Info	Stock	Sovereign
15 Oct 2008 (1)	ECB says list of assets to eligible as collateral in Eurosystem credit operations will be expanded until the end of 2008	In its latest bid to tame the financial market crisis, the ECB lowered its threshold for accepting assets to BBB- from A- and said it would accept assets denominated in foreign currencies and debt instruments issued by credit institutions, among other changes. The dollar and euro both pared losses against the yen on Wednesday after the European Central Bank said it would extend a series of efforts to boost liquidity until the end of 2009. The moves, which include a plan to offer U.S. dollar liquidity through foreign currency swaps, eased risk aversion. Money market interest rates have eased this week as banks now have fewer concerns about access to liquidity and analysts said the changes should help to bring wholesale rates down further.	(-)	1
26 Nov 2008 (2)	ECB says to no longer accept syndicated loans as collateral	No relevant market commentary.	(+)	0

Date	Title	Market Commentary/Background Info	Stock	Sovereign
20 Jan 2009 (3)	ECB says to require a rating of AAA/Aaa for ABS as collateral issued as of March 1	Analysts were mixed in their reaction to the move, the third substantive change to the ECB's collateral framework since last September, as the financial crisis has intensified. "From the banks' perspective, it's more restrictive, but it's not so drastic that it will shrink the mortgage market," said an analyst specialising in structured credit at a German bank. RBS economist Silvio Peruzzo said the announcement was minor compared to other changes announced to the ECB's collateral rules. "It provides an additional control on the quality of assets," he said. But Tullett Prebon economist Lena Komileva said the move sent the wrong signal at a time when the Bank of England and the U.S. Federal Reserve were stepping in to support illiquid markets. "Rather than responding to an underlying improvement in market liquidity conditions, the ECB's collateral tightening reflects quite the opposite – poor asset quality and insufficient market transparency – resulting in greater financial stresses," she said.	(-)	0
03 May 2010 (4)	ECB suspends limits on Greek debt	Move avoids Greek debt falling off collateral list * Removes role of rating agencies in Greek crisis * They want to make sure that the question of Greek banks having access to liquidity will not be an issue in this crisis. It's one less thing to worry about." "We always thought the ECB would be quite flexible and pragmatic.	(-)	1*
28 Jul 2010 (5)	ECB lays out new sliding-scale of collateral haircuts	ECB tightens rules on weaker-rated collateral. The changes are slightly more aggressive than expected, but overall they should not impact banks financing too significantly, said Credit Agricole economist Frederik Ducrozet. "The biggest changes are on lower-rated assets and they are the type of assets that are not used to such a large degree."	(-)	1

Date	Title	Market Commentary/Background Info	Stock	Sovereign
09 Oct 2010 (6)	ECB underscores power to limit banks' borrowing	The changes, published on Saturday, clarify the ECB's ability to bar or restrict banks from borrowing from the ECB and impose ad-hoc limits on what assets can be swapped for loans. "On the grounds of prudence, the Eurosystem may also reject assets, limit the use of assets or apply supplementary haircuts to assets submitted as collateral in Eurosystem credit operations by specific counterparties."	(+)	0
16 Dec 2010 (7)	ECB says it plans to introduce loan-by-loan information requirements for ABS in collateral framework within 18 months	News mainly about capital increase ECB. "We infer from this that the ECB ... is seeking a greater cushion in order to offset potential losses, given that its portfolio of securities holdings has risen substantially, as well as to protect itself from potential collateral losses," Barclays Capital economists said in a research note.	(-)	0
31 Mar 2011 (8)	ECB has decided to suspend the application of minimum credit rating threshold in collateral eligibility requirements for Ireland government bonds until further notice	* ECB helps Irish banks by loosening lending requirements * Commits to continue providing funds * But appears to shelve plan for new liquidity facility * Disagreements over plan within Governing Council.	(+)	1*
29 Apr 2011 (9)	ECB to require loan-by-loan data for commercial MBSs	No relevant market commentary.	(-)	0

Date	Title	Market Commentary/Background Info	Stock	Sovereign
21 Sep 2011 (10)	ECB says Eurosystem has reduced limit for use of unsecured debt instrument issued by banks, such assets may be used as ECB collateral with value not to exceed 5 pct vs 10 pct earlier	ECB abolishes eligibility rules for some bank-issued debt * Reduces limit on bank-issued unsecured debt to 5 pct * Analysts see net loosening effect. The move comes amid growing fears of a renewed credit crunch as the euro zone sovereign debt crisis is shaking confidence in the region's banks. "They are really trying to make it easier for banks to access liquidity, widening the pool of assets they can use," Marie Diron, an economist at Oxford Economics, said. "They are trying to avoid problems in the sector which would then spill over to the economy." The 17-country bloc's central bank also said it had limited the use of bank-issued unsecured debt instruments. "Such assets may only be used as collateral to the extent that the value assigned does not exceed 5 percent of the total value of collateral submitted," the ECB said. The limit was previously 10 percent. "I'm not sure that's too much of a constraint, as big banks have not been able to issue many of these bonds lately, so it was not their main source of collateral," Diron said.	(+)	0*
28 Feb 2012 (11)	ECB has decided to temporarily suspend eligibility of Greek bonds used as collateral	A senior Greek banker said the only access for banks during this period will be the Bank of Greece, which will expand the ELA facility to accommodate liquidity needs until the European Financial Stability Facility money is available. That the euro zone's deal makers and the ECB did not foresee the potential risk of the EFSF support scheme not being activated in time is likely to leave some with red faces. The issue is vital because Greek banks would almost certainly go bust if their central bank funding was withdrawn. Other banks in countries like France also own large chunks of Greek debt, though they have other assets to use as collateral. "The decision of the ECB to suspend temporarily Greek bonds as collateral has no impact on French banks," a Bank of France spokeswoman said.	(+)	1*

Date	Title	Market Commentary/Background Info	Stock	Sovereign
22 Jun 2012 (12)	ECB reduces rating threshold for ABS	German debt extended falls and stocks trimmed losses on Friday after the European Central Bank announced it would accept a wider range of assets as collateral, including those of a lower quality, in move designed to ease pressure on Spain. [ID:nF9E8F402O] "It just means that they are willing to take as collateral lower-quality credit, which is probably why the periphery is getting a bit of a bid against Germany," a trader said.	(+)	0*
03 Jul 2012 (13)	ECB says banks may not submit government guaranteed bank bonds as collateral above nominal value without ECB Pre-approval	No relevant market commentary. Background info: banks in troubled euro zone countries such as Greece have been increasingly borrowing ultra-cheap funds from the ECB using self-issued bonds which are then given a state guarantee by the government which make them eligible at the ECB. The amounts involved could be over 100 billion euros. Central bank policymakers have become increasingly disgruntled by the practice in recent months, worried it could leave the ECB facing enormous losses if Greece or others doing the same left the euro.	(+)	0*
06 Jul 2012 (14)	ECB to require more data on ABS starting this year	No relevant market commentary.	(-)	0

Date	Title	Market Commentary/Background Info	Stock	Sovereign
20 Jul 2012 (15)	ECB says Greek government bonds to be ineligible as collateral from July 25	Greek banks would almost certainly go bust if their central bank funding was withdrawn. Banks in other euro zone countries also own large chunks of Greek debt, though they are more likely have other assets to use as collateral and are thus not hit as hard. "In this way the ECB could be putting pressure (on the Greek government) to bring about a positive review by the troika," Alpha Finance bank analyst Nikos Lianeris said. "If there is a positive review by the troika then the Greek banks will regain direct access to ECB funding." Greek bankers took the decision in their stride. "It's something we were expecting," one banker speaking on the condition of anonymity said. "The only difference is the borrowing cost for the banks." In a separate statement, the ECB said it would start accepting some Greek credit claims as collateral, but this move is unlikely to make up for much of the exclusion of the country's sovereign bonds. The ECB also said it had agreed to extend the collateral use of credit claims by banks in struggling euro zone members Cyprus, Portugal and Italy. The ECB could not provide immediate estimates of how much extra collateral the changes would provide for banks. It also could not say why Spain, whose ailing banks will receive a bailout, was not included in the expansion.	(-)	1*
27 Nov 2012 (16)	ECB delays implementation of ABS loan data reporting	No relevant market commentary.	(-)	0

Date	Title	Market Commentary/Background Info	Stock	Sovereign
19 Dec 2012 (17)	ECB - Greek sovereign debt shall again constitute eligible collateral for ECB credit operations, subject to haircuts	Lifting a ban in place since July, the ECB said it was making Greek debt eligible again in light of the country's progress with reform measures, budget cutting and privatisations. Greek bank shares extended gains on the news, some trading up more than 10 percent on the day. Greek bonds rallied sharply in response to the change, which makes the debt more attractive for domestic banks, with 10-year sovereign yields falling by almost 1.5 percentage points to 11.42 percent.	(+)	1*
22 Mar 2013 (18)	ECB says it will stop collateral use of uncovered government-guaranteed bank bonds that have been issued by bank itself, or entity closely linked to it, from March 1, 2015	No relevant market commentary.	(-)	0*
28 Jun 2013 (19)	ECB says it has decided to temporarily suspend eligibility of Cyprus debt for use as collateral at its monetary policy operations	Background: In response to Cyprus's announcement on the debt exchange, Standard & Poor's downgraded the sovereign foreign currency credit rating on Cyprus to SD (selective default) from CCC. S&P said that after the exchange, which is expected to occur on July 1, the liquidity strains on the government should be alleviated. After the exchange, the rating is expected to rise to CCC-plus. The ECB move is likely to be temporary.	(+)	1*
05 Jul 2013 (20)	ECB lifts suspension on Cyprus debt eligibility	No relevant market commentary. Background info: The European Central Bank said on Friday it was lifting a suspension on the eligibility of Cyprus's debt for use in its refinancing operations after Standard & Poor's upgraded the island state's credit rating."	(+)	1*

Date	Title	Market Commentary/Background Info	Stock	Sovereign
18 Jul 2013 (21)	ECB says to expand list of collateral it accepts at liquidity operations	*Banks rally as ECB expands collateral rules* French, Italian and Spanish banks extend gains, with sentiment on the sector bolstered after the European Central Bank expands the list of eligible collateral to include more asset-backed securities (ABS). Among the top risers are Popolare Milano (PMIL.MI) and Bankinter (BKT.MC), which add 8.5 percent each. "Shares in Italian banks are boosted by the ECB move because they could be among the main beneficiaries of these new rules as they have high exposure" to loans for SMEs, says a trader in Milan.	(+)	0*
09 Sep 2013 (22)	ECB says it could offer leniency in ABS loan reporting	Background info: this move offers some exceptions to the ECB's loan-level reporting requirements, which was designed to increase transparency for the hard-to-value assets.	(+)	0
19 Sep 2013 (23)	ECB asks for data on ABS backed by credit card debt	No relevant market commentary.	(-)	0
27 Sep 2013 (24)	ECB collateral rules to increase ABS eligibility apply from Oct	Background info: The European Central Bank's new collateral rules that let banks use more of the assets once blamed for triggering the financial crisis in the central bank's refinancing operations will come into force in October, the ECB said on Friday.	(+)	0
24 Jan 2014 (25)	ECB postpones minimum threshold for credit claims	No relevant market commentary.	(-)	0
12 Mar 2014 (26)	ECB's Coeure - Development of ABS market is probably beyond remit of monetary policy, European Commission should lead it	No relevant market commentary.	(-)	0

Date	Title	Market Commentary/Background Info	Stock	Sovereign
19 Sep 2014 (27)	ECB's Praet urges governments to re-think support for ABS	The ECB will detail its plans at its October policy meeting. Some analysts have criticised the plan, saying it would fail to free capital on banks' balance sheets if the ECB sticks to only buying the senior and most secure ABS tranches, leaving the riskier parts with the bank.	(-)	0
04 Feb 2015 (28)	ECB lifts waiver on credit rating requirements for Greek bonds	U.S. Treasuries prices turned positive in late trading on Thursday, erasing earlier losses after the European Central Bank would not accept Greek government bonds as collateral for loans to banks. The ECB move rekindled safe-haven demand for low-risk U.S. and German government debt as it stoked worries about runs at Greek banks and concerns the new Greek government would have trouble renegotiating debt terms with its euro zone partners, analysts said.	(-)	1*
20 Feb 2015 (29)	ECB ready to reintroduce waiver for Greek collateral once it assesses that program likely to be concluded - ECB sources	No relevant market commentary.	(+)	1*
31 Aug 2015 (30)	ECB to accept loan bundles as collateral	The decision is aimed at expanding the pool of collateral available to euro zone banks and stimulating lending, which has been slow as the economy struggles.	(+)	0
22 Jun 2016 (31)	ECB set to reinstate Greek banks' access to cheap funding	The reinstatement of the waiver for Greek assets by the ECB could benefit Greek core banks' net interest income by as much as around 80 million euros – after tax at around 60 million euros – depending on the eligibility of the Greek assets and the level of haircut imposed, Greek brokerage Euroxx said. "This, in our view, will be essential for the reduction of Greek banks funding costs, which along with the gradual easing of capital controls should also help the all-important return of deposits into the system," it said.	(+)	1*

Date	Title	Market Commentary/Background Info	Stock	Sovereign
05 Oct 2016 (32)	ECB says it has decided to reduce, as of 1 Jan 2017, the usage limit for uncovered bank bonds from 5% to 2.5% for collateral eligibility	Background info: The European Central Bank will continue to accept unsecured bank bonds as collateral for lending, including some that may be written down if a bank fails, but it is restricting their use and increasing checks, it said on Wednesday. The move comes in response to new European rules stating that investors in a bank, including holders of bonds that are not backed by collateral, must suffer losses in case of default before public money can be used.	(+)	0*
03 Nov 2016 (33)	ECB to tighten collateral rules	No relevant market commentary.	(+)	0
14 Dec 2017 (34)	ECB says UBSS that are subject to statutory, contractual or structural subordination (e.g., UBSS issued by bankholding companies) will become ineligible as collateral	No relevant market commentary.	(-)	0
08 Feb 2018 (35)	ECB says it amends criteria on interest payment structures for eligible credit claims and other technical changes related to collateral framework	No relevant market commentary. Background info: The European Central Bank is tightening regulations to prevent banks from pledging some riskier forms of collateral when they borrow from euro zone central bank.	(-)	0

Date	Title	Market Commentary/Background Info	Stock	Sovereign
22 Mar 2019 (36)	ECB says eligibility requirements for loan-level data reporting in collateral framework to be adjusted to reflect EU securitisation regulations and disclosure requirements	No relevant market commentary.	(-)	0
07 Apr 2020 (37)	ECB announces package of temporary collateral easing measures	No relevant market commentary. Background info: In a series of measures unveiled after an unscheduled Governing Council meeting, the ECB said it would ease collateral requirements, making it easier for banks to use its funding operations, which banks tapped for hundreds of billions of euros.	(-)	1
22 Apr 2020 (38)	ECB says it takes steps to mitigate impact of possible rating downgrades on collateral availability	Background info: Limited reaction to ECB, market focused on EU conference.	(+)	1
22 Sep 2020 (39)	ECB says to accept sustainability-linked bonds as collateral	Nevertheless, analysts expect that ECB announcement to trigger an issuance boom.	(+)	0

Date	Title	Market Commentary/Background Info	Stock	Sovereign
08 Jul 2021 (40)	ECB says it will introduce disclosure requirements for private sector assets as a new eligibility criterion or as a basis for a differentiated treatment for collateral and asset purchases	"Other central banks are going to be reading this and thinking hard about how they can show a similar commitment to greening monetary policy," said Paul Diggle, deputy chief economist of Aberdeen Standard Investments.	(+)	0
24 Mar 2022 (41)	ECB to gradually phase out collateral-easing measures between Jul 2022 and Mar 2024	No relevant market commentary. Background info: "The ECB also reaffirmed a waiver on Greek government bonds for as long as it keeps investing the proceeds from its Pandemic Emergency Purchase Program (PEPP). In a hopeful sign for other countries with lower credit ratings, such as Cyprus, Portugal or Italy, the ECB added that it may disregard agencies' ratings again in the future." "The ECB's Governing Council reserves the right to deviate also in the future from credit rating agencies' ratings if warranted," the ECB said.	(-)	1
04 Jul 2022 (42)	ECB says it will limit the share of assets issued by high polluters that can be pledged as collateral when borrowing from the ECB	No relevant market commentary.	(+)	0
02 Dec 2022 (43)	ECB says Eurosystem reschedules launch of new collateral management system	No relevant market commentary.	(+)	0

Date	Title	Market Commentary/Background Info	Stock	Sovereign
20 Dec 2022 (44)	ECB tightens collateral rules. Bonds issued by the European Commission receive an haircut category upgrade	No relevant market commentary.	(+)	1

A.2 Collateral Policy Transmission in the Model

Our stylized framework presented in Section 2.2, can also be used to study how collateral policy changes affect the pricing of eligible assets. In the remainder of this section, we interpret the assets held by banks as government bonds and solve the shareholder value maximization problem, which yields a pricing condition for government bonds q_0^B . We assume that bank liabilities d_1 are priced competitively at their expected repayment probability, such that we can write the maximization problem as:

$$\max_{d_1, b_1} \mathbb{E}_0[\text{div}_1] \quad \text{s.t.} \quad \text{div}_0 = 0 \quad \text{and} \quad q_0^D = 1 - D^{\text{bank}}(\bar{b}_1/d_1). \quad (\text{A.1})$$

After some re-arrangements¹² the first-order conditions with respect to b_1 are given by

$$q_0^B = \left(\underbrace{F(\bar{b}_1/d_1)}_{\text{Discounting}} + \underbrace{\frac{\bar{b}_1}{d_1} f(\bar{b}_1/d_1)}_{\text{Coll. premium}} \right) \cdot \mathcal{R}_1. \quad (\text{A.2})$$

The possibility of bank failure, a higher $F(\bar{b}_1/d_1)$, affects asset prices in two ways. First, the expected payoff from purchasing *any* asset is discounted more strongly since banks receive the payoff from their investment in fewer states of the world. Second, holding pledgeable securities reduces the bank default probability, since they increase the likelihood that the collateral value of their assets is sufficient to cover the liquidity shock $\delta_1 d_1$. We refer to this willingness to pay for collateral services as the *collateral premium*, which in turn depends on the amount of collateral services provided by the asset $\frac{\bar{b}_1}{d_1}$, and the marginal default probability $f(\bar{b}_1/d_1)$.

How does collateral policy affect the government bond price? Since collateral policy does not affect each asset's expected payoff $\bar{R}_1$, we can exploit the exponential distributional assumption on the liquidity deficit and write the partial derivatives of the asset pricing condition with respect to the collateral parameter κ :

$$\frac{\partial q_0}{\partial \kappa} = \left(\underbrace{-f(\bar{b}_1/d_1)}_{\text{More bank failure}} - \underbrace{f(\bar{b}_1/d_1)}_{\text{Smaller coll. amount}} - \underbrace{\frac{\bar{b}_1}{d_1} f'(\bar{b}_1/d_1)}_{\text{Larger coll. valuation}} \right) \cdot \frac{b_1 \mathcal{R}_1^2}{d_1}, \quad (\text{A.3})$$

A collateral policy tightening (higher κ) depresses banks' valuation of government bonds by raising expected bank failure. At the same time, the collateral service provided by the affected government bond is smaller. However, this also makes the remaining collateral services provided

¹²Using equations (1) and (3), the objective function can be written as

$$\max_{d_1, b_1} \left(1 - D^{\text{bank}}(\bar{b}_1/d_1) \right) (\mathcal{R}_1 b_1 - d_1) \quad \text{s.t.} \quad \left(1 - D^{\text{bank}}(\bar{b}_1/d_1) \right) d_1 = q_0 b_1.$$

Plugging in the constraint, this can be expressed as

$$\max_{b_1} \left(1 - D^{\text{bank}}(\bar{b}_1/d_1) \right) \mathcal{R}_1 b_1 - q_0^B b_1.$$

by the bond more valuable. Kaldorf and Roettger (2025) provide a discussion of these two opposing channels in the context of convenience yield on risky euro area government debt. Which channel dominates or whether they approximately offset each other is hence a quantitative question.

To understand the transmission of collateral policy to sovereign CDS, it is helpful to note that we can price an asset that reflects the payoff profile of a sovereign CDS in our stylized model:

$$q_0^{CDS} = F(\bar{b}_1/d_1) \mathcal{R}_1^{CDS}. \quad (\text{A.4})$$

While the CDS price is also negatively affected by the bank failure rate, it does not contain the second term related to the *marginal* bank failure rate, since a CDS is an insurance against sovereign default and is not pledgeable as collateral. The partial derivative of the CDS price with respect to the collateral parameter reads

$$\frac{\partial q_0^{CDS}}{\partial \kappa} = -f(\bar{b}_1/d_1) \frac{\mathcal{R}_1 b_1}{d_1} \mathcal{R}_1^{CDS}. \quad (\text{A.5})$$

Intuitively, a lower bank failure rate increases banks' willingness to pay for protection against a sovereign default event, since they receive the payoff from such an asset with a larger probability. The model also predicts a larger response of government bond spreads and CDS spreads if these assets are held and priced by riskier banks, since the marginal effect of collateral availability on bank failure is larger. We test these predictions when estimating the effects of our identified collateral policy surprise series on a panel of government bond and CDS spreads. We provide supporting evidence for this mechanism by comparing the capitalization, liquidity proxies, and the share of domestic government bonds for banks located in euro area core and periphery countries.

Model Prediction IIa. Expansionary collateral policy lowers government bond and CDS spreads over the risk-free rate.

Model Prediction IIb. The effects are stronger for bonds held and CDS priced by riskier banks.

A.3 Bank Selection and Bank Heterogeneity

Table A.2 demonstrates that removing one bank at a time does not affect the construction of the collateral policy surprise series. Specifically, the first column demonstrates that the correlation is at least 99.5%. The second column shows how the average stock price reaction changes when individual institutions are left out. A positive Δ *mean reaction* implies that the omitted bank profits relatively less from a collateral policy expansion. As the last column shows, the exact bank selection is also less relevant when it comes to classifying individual events into contractionary or expansionary.

Bank	Correlation	Δ Mean Reaction	# Sign Flips
BBVA	0.998	0.020	1
BC Portugues	0.997	-0.013	4
BNP Paribas	0.997	0.003	1
Deutsche Bank	0.995	0.022	0
Intesa Sanpaolo	0.998	0.008	2
Santander	0.996	0.018	0
Societe Generale	0.996	0.005	1
UniCredit	0.997	0.011	1

Table A.2: "Leave-One-Out" Surprise Series

In Figure A.1, we compare how individual banks respond to each announcement, relative to the mean stock price reaction. While the blue dots correspond to our baseline collateral policy surprises, the red dots display each individual bank's response. Note that high-frequency stock price data are not consistently available for Commerzbank and ING throughout the entire sample, such that we focus on the remaining eight banks in Table A.2 and Figure A.1.

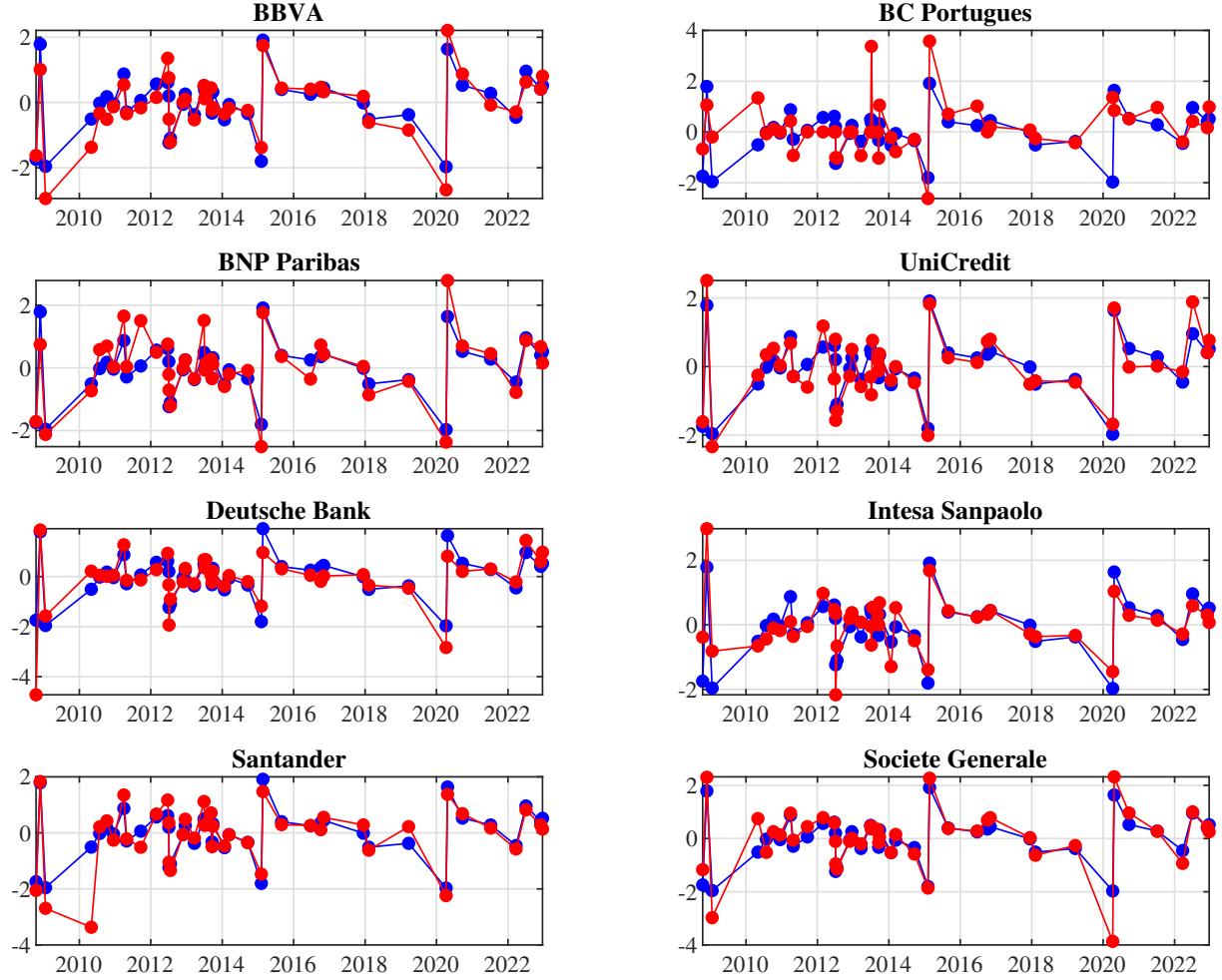


Figure A.1: Collateral Policy Surprises: Individual Stock Price Reactions. This figure plots the stock price response by each individual bank against the average stock price reaction. Note that the average response is different from, but highly correlated with, the first principal component that we use as collateral policy surprise.

In Figure A.2, we demonstrate that the distinction between banks located in euro area core and periphery countries does not affect the *measurement* of collateral policy surprises over a short time window. We show that the first principal component in either sub-sample is very highly correlated with the first principal component in the full sample of banks.

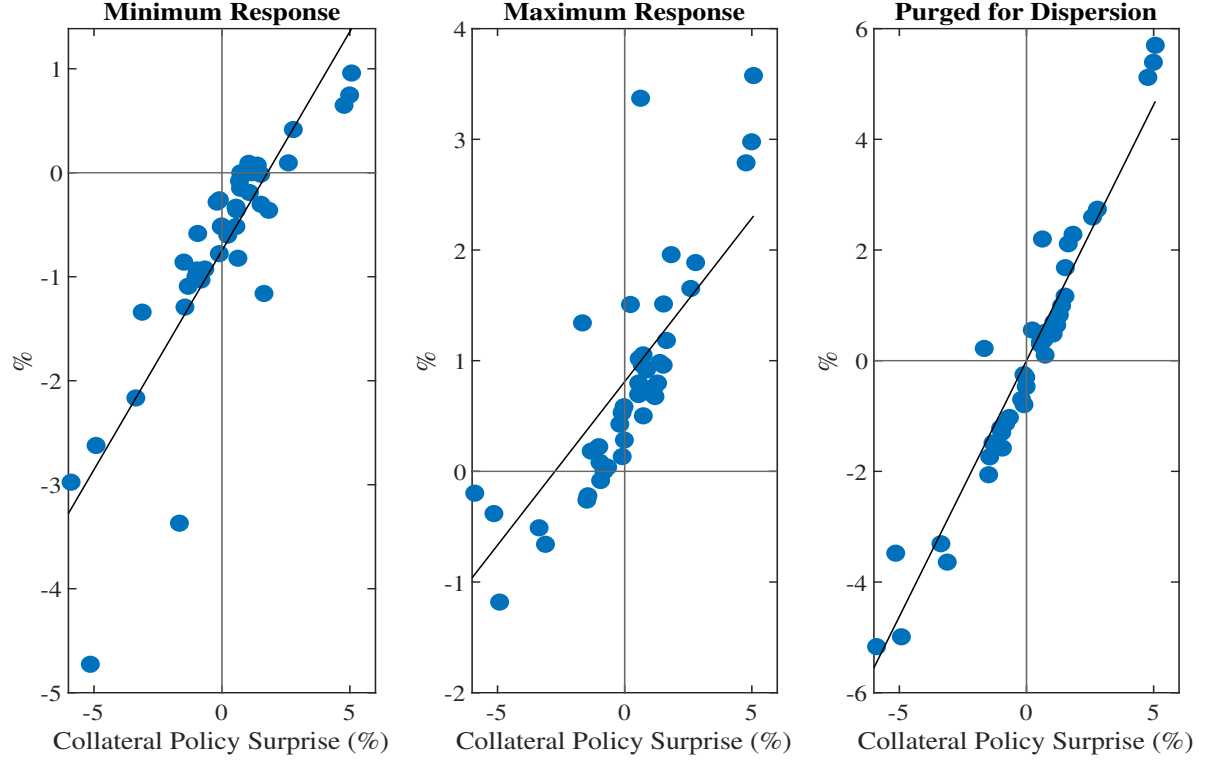


Figure A.2: Collateral Policy Surprises and the Role of Bank Selection. The top row compares our baseline collateral policy surprise to the surprises estimated on the subsample of core and periphery banks, respectively. The bottom row compares our baseline collateral policy surprise to the maximum and minimum stock price reaction at each event. The solid line represents a simple linear fit.

Table A.3 presents summary statistics of all bank characteristics that we use as control variables when estimating the determinants of the stock price reaction in equation (8). By construction, all considered banks are quite similar in terms of size, but exhibit otherwise very different characteristics. For example, the average stock return varies from -2% for the worst-performing bank to 8% for the best-performing. The average tier 1 equity ratio varies from 7% for one bank in one year, which is very close to the regulatory minimum, to more than 19%.

Variable	Mean	SD	Min	Max	N
Δ Stockreturn	0.29	1.42	-2.94	8.10	235
ln(Assets)	13.49	0.94	11.18	14.92	235
Loans Ratio	51.18	15.22	12.74	80.49	235
ROE	6.05	4.22	0.00	15.62	235
Debt Ratio	25.94	12.05	2.29	56.22	235
Deposits Ratio	52.58	15.73	19.87	81.88	235
Leverage Ratio	25.94	12.05	2.29	56.22	235
Sec over Assets	33.54	12.99	10.44	75.63	235
Tier1Ratio	13.19	2.62	7.10	19.10	216

Table A.3: Cross Section of Banks: Descriptive Statistics.

B Additional Results: Transmission

This section presents additional results on the transmission of collateral policy. We show the responses of individual bank CDS spreads and governments in Appendix B.1 and the CDS-bond basis in Appendix B.2. Appendix B.3 contains additional results with respect to pre-trends and orthogonalized collateral policy surprises. We also provide a discussion of the international transmission (Appendix B.4) and show effects at the monthly frequency (Appendix B.5).

B.1 Banks and Government Bond Market

The results presented in this section complement the government bond market effects of collateral policy surprises in Section 4.2, which were based on splitting countries and banks into the euro area core and periphery.

Individual Government Bond Spreads In Figure B.1, we display the response of individual countries to an expansionary CPS. Consistent with the panel local projections presented in the main text, we find little or no effect on core countries, while all periphery spreads decline by 5 to 10 basis points within two days after the event.

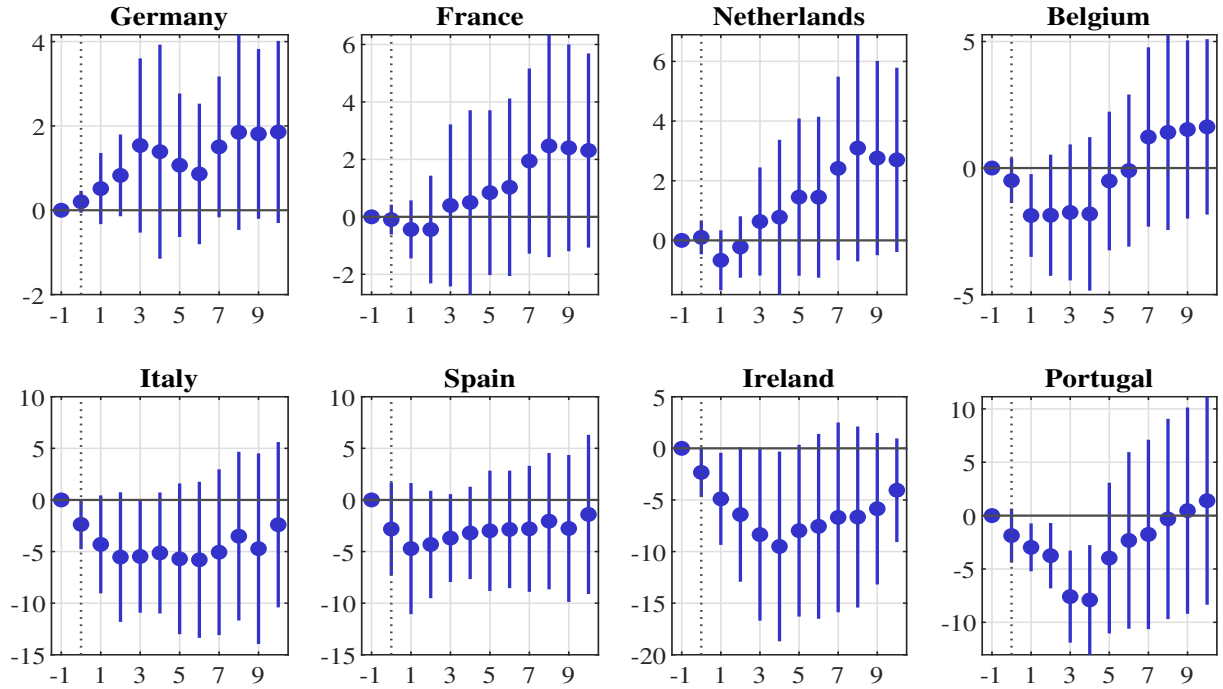


Figure B.1: Collateral Policy Surprises and 5Y Bond Spreads over OIS: Results from estimating (10) for 5-year government bond yield spreads relative to OIS for all countries. Dots represent estimated coefficients, and vertical lines denote 90% confidence intervals based on heteroskedasticity- and autocorrelation-consistent (Newey-West) standard errors. The horizontal axis reports event time in days relative to the policy announcements, with $t = 0$ marking the announcement dates.

Individual Sovereign CDS In Figure B.2, we present the response of individual sovereign CDS spreads. The upper row shows that German and French CDS spreads decline by about 2bps, Italian and Spanish CDS spreads peak at around -8bps, which is remarkably similar to the effects on government bond spreads.

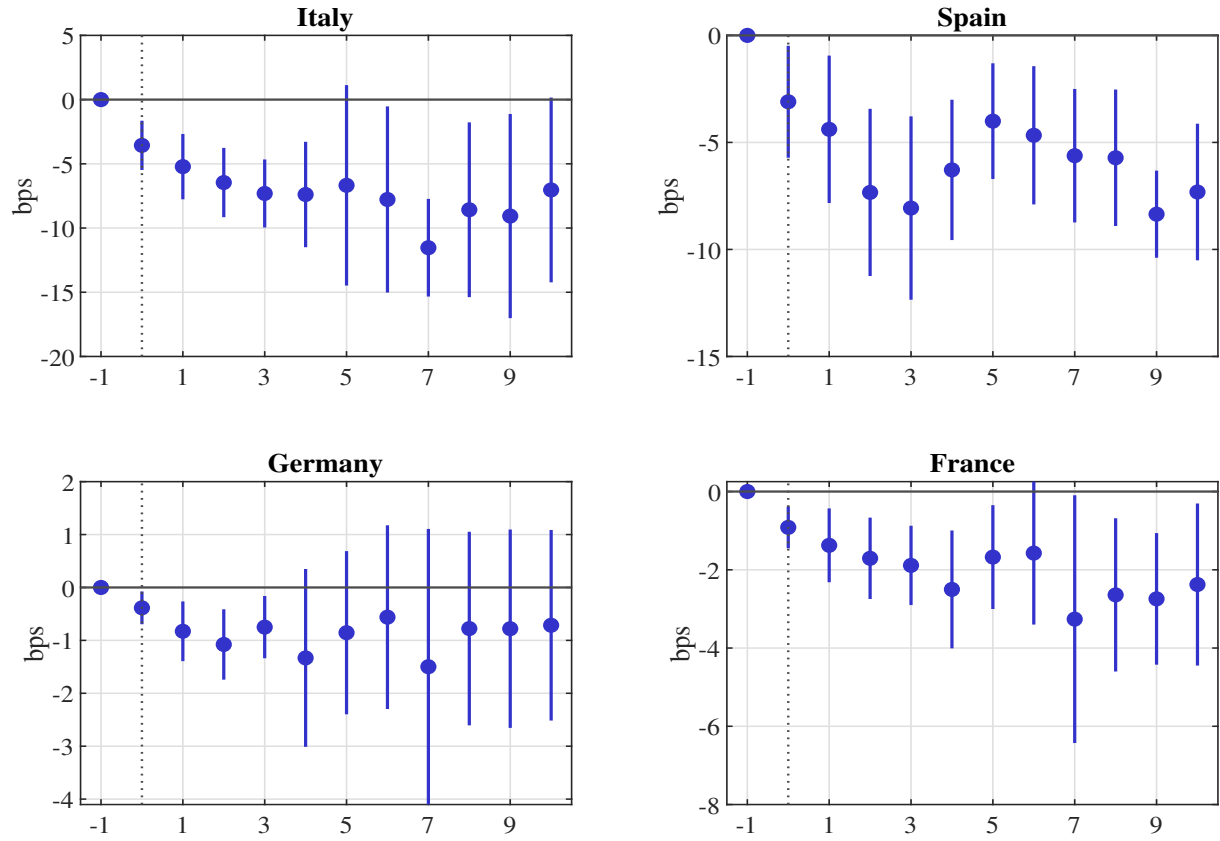


Figure B.2: Collateral Policy Surprises and Sovereign CDS: Results from estimating equation (9) for 5-year Sovereign Credit Default Swap Spreads. Dots represent estimated coefficients and vertical lines denote 90% confidence intervals based on heteroskedasticity- and autocorrelation-consistent (Newey-West) standard errors. The horizontal axis reports event time in days relative to the policy announcements, with $t = 0$ marking the announcement dates.

Individual Bank CDS In Figure B.3, we complement our results from the main text by plotting the response of individual bank CDS spreads. It is worth noting that the response is persistently negative for each individual bank and economically meaningful. Pre-announcement trends turn out to be generally small and insignificant for all banks. The banks with the largest responses are all located in the periphery (Santander, Unicredit, Intesa Sanpaolo and BBVA). For most of the core banks, the CDS drops significantly as well, but the magnitude is less than half than for periphery banks and the effects are generally less significant.

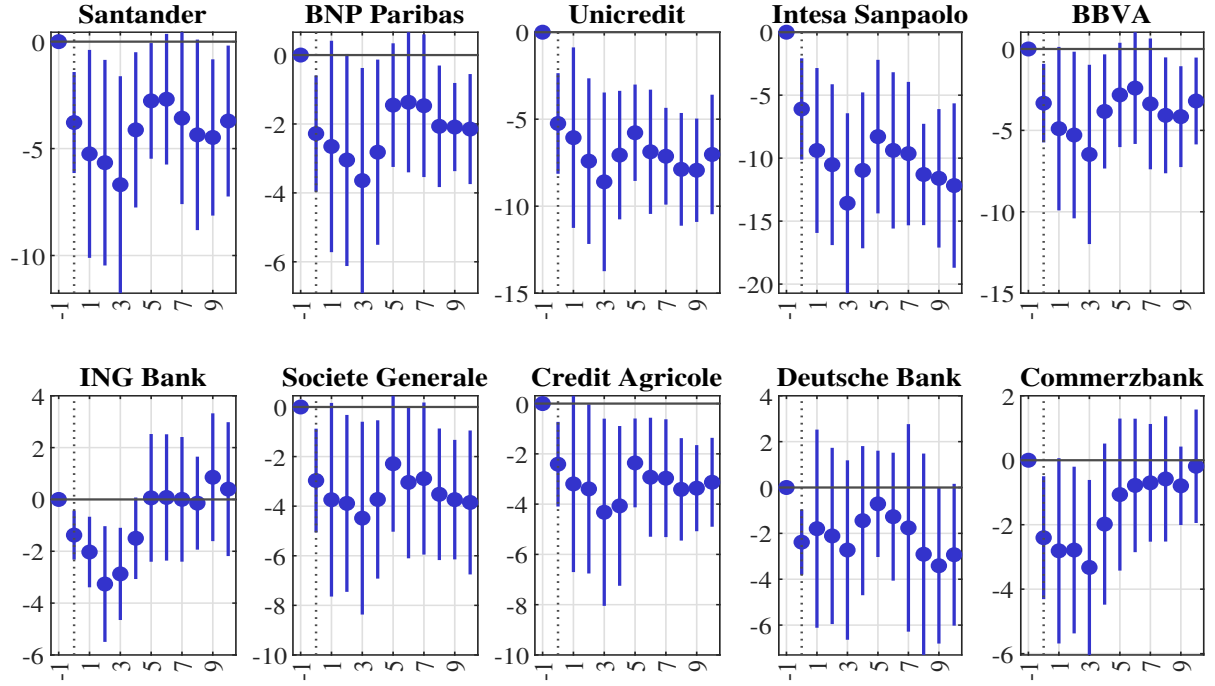


Figure B.3: Bank CDS: Results from estimating equation (10) using all collateral policy surprises. The dependent variables are 5 Year senior CDS written on single-name banks, expressed in basis points. Dots represent estimated coefficients, and vertical lines denote 90% confidence intervals based on heteroskedasticity- and autocorrelation-consistent (Newey-West) standard errors. The horizontal axis reports event time in days relative to the policy announcements, with $t = 0$ marking the announcement dates.

B.2 CDS-Bond Basis

In Figure B.4, we show that the CDS-bond basis of country i , which we define as $basis_{i,t} \equiv ois_t - cds_{i,t} - yield_{i,t}$ does not respond to a collateral policy surprise. The basis is positive for all core countries and for periphery countries outside of the euro area debt crisis, indicating that governments could issue debt at comparatively low yields. While collateral eligibility is one common microfoundation for the persistent discrepancy between government bond yields and the combination of risk-free rate and CDS-spread, our findings indicate that collateral policy has little impact on the CDS-bond basis.

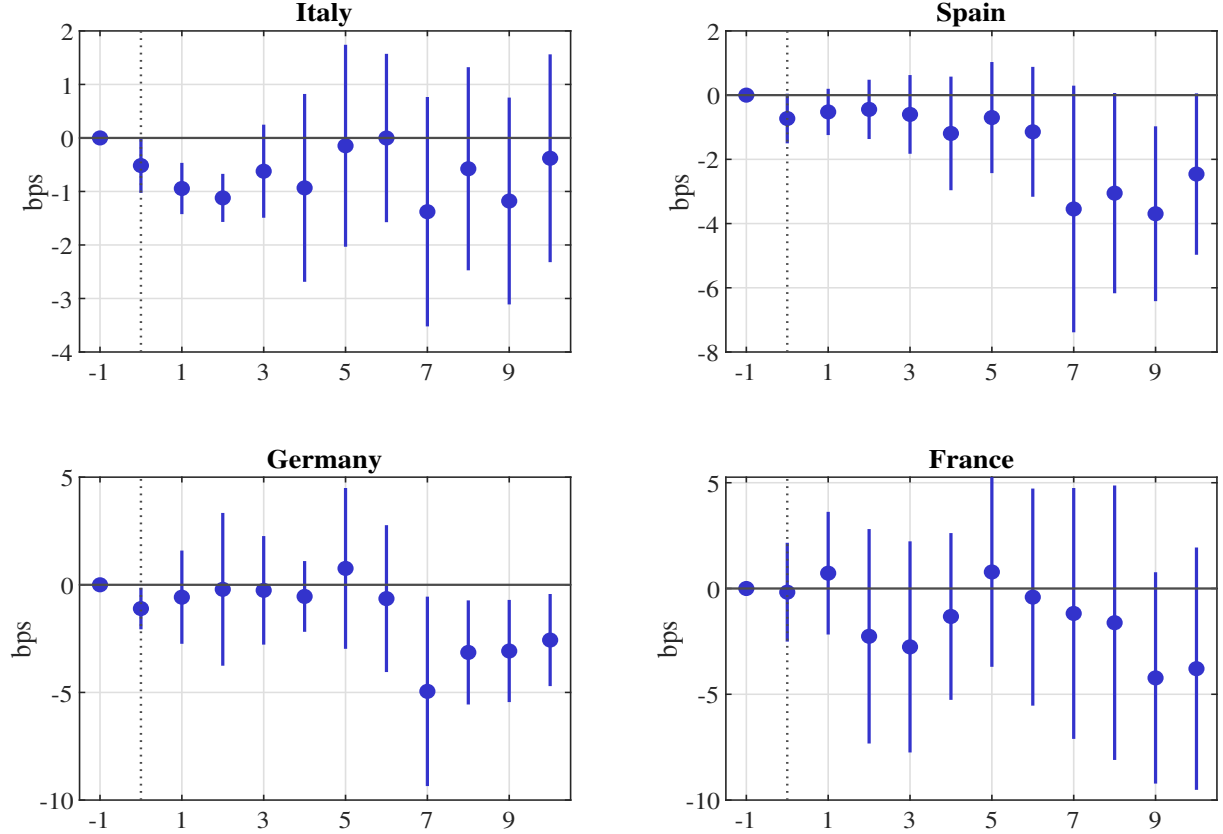


Figure B.4: CDS-Bond Basis: Results from estimating (10) for the 5-year CDS-Bond Basis. Dots represent estimated coefficients, and vertical lines denote 90% confidence intervals based on heteroskedasticity- and autocorrelation-consistent (Newey-West) standard errors. The horizontal axis reports event time in days relative to the policy announcements, with $t = 0$ marking the announcement dates.

B.3 Sensitivity checks for pre-trends and orthogonalized CPS

As a first sensitivity check, we expand the estimation horizon by three trading days before the policy change. In Figure B.5, we report the results for financial market variables, while Figure B.7 contains the results for the government bond market and bank CDS spreads. In both cases, there are no significant pre-trends. As a second sensitivity check, we orthogonalize the baseline collateral policy surprises using the most powerful set of potential predictors discussed in Table 1. Figure B.6 and Figure B.8 demonstrate that there are no visible differences to our baseline specification presented in the main text.

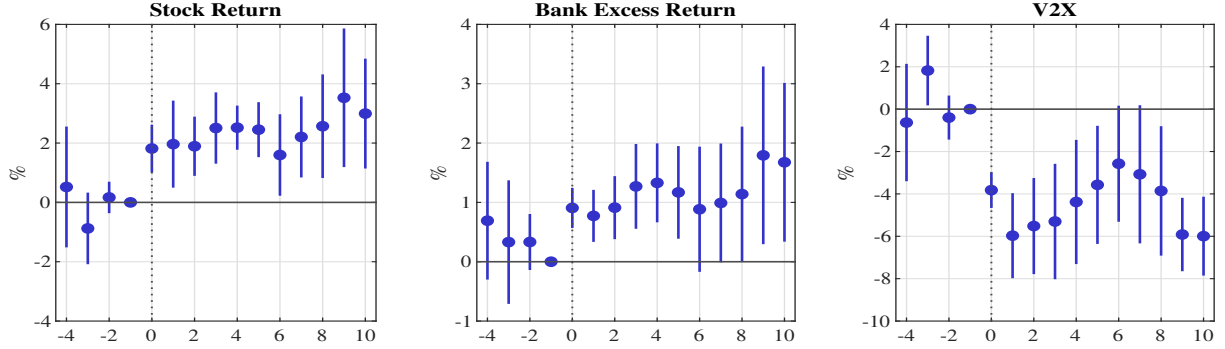


Figure B.5: Pre-trend sensitivity check to key financial indicators: Results from estimating equation (9) using all collateral policy surprises for $h = -4$ to $h = 10$. The dependent variables are the EuroSTOXX50 return, the excess return of EuroSTOXX Bank Index over the overall index, and the implied volatility of EuroSTOXX50 options (V2X). Dots represent estimated coefficients, and vertical lines denote 90% confidence intervals based on heteroskedasticity- and autocorrelation-consistent (Newey-West) standard errors. The horizontal axis reports event time in days relative to the policy announcements, with $t = 0$ marking the announcement dates.



Figure B.6: Orthogonalized CPS sensitivity check to key financial indicators: Results from re-estimating equation (9) using collateral policy surprises orthogonalized with respect to the potential predictors included in specification (3) of Table 1. Following Bauer and Swanson (2023), the orthogonalized surprise series is constructed as the residual from regressing the baseline collateral policy surprises on these predictors. The dependent variables are the EuroSTOXX50 return, the excess return of EuroSTOXX Bank Index over the overall index, and the implied volatility of EuroSTOXX50 options (V2X). Dots represent estimated coefficients, and vertical lines denote 90% confidence intervals based on heteroskedasticity- and autocorrelation-consistent (Newey-West) standard errors. The horizontal axis reports event time in days relative to the policy announcements, with $t = 0$ marking the announcement dates.

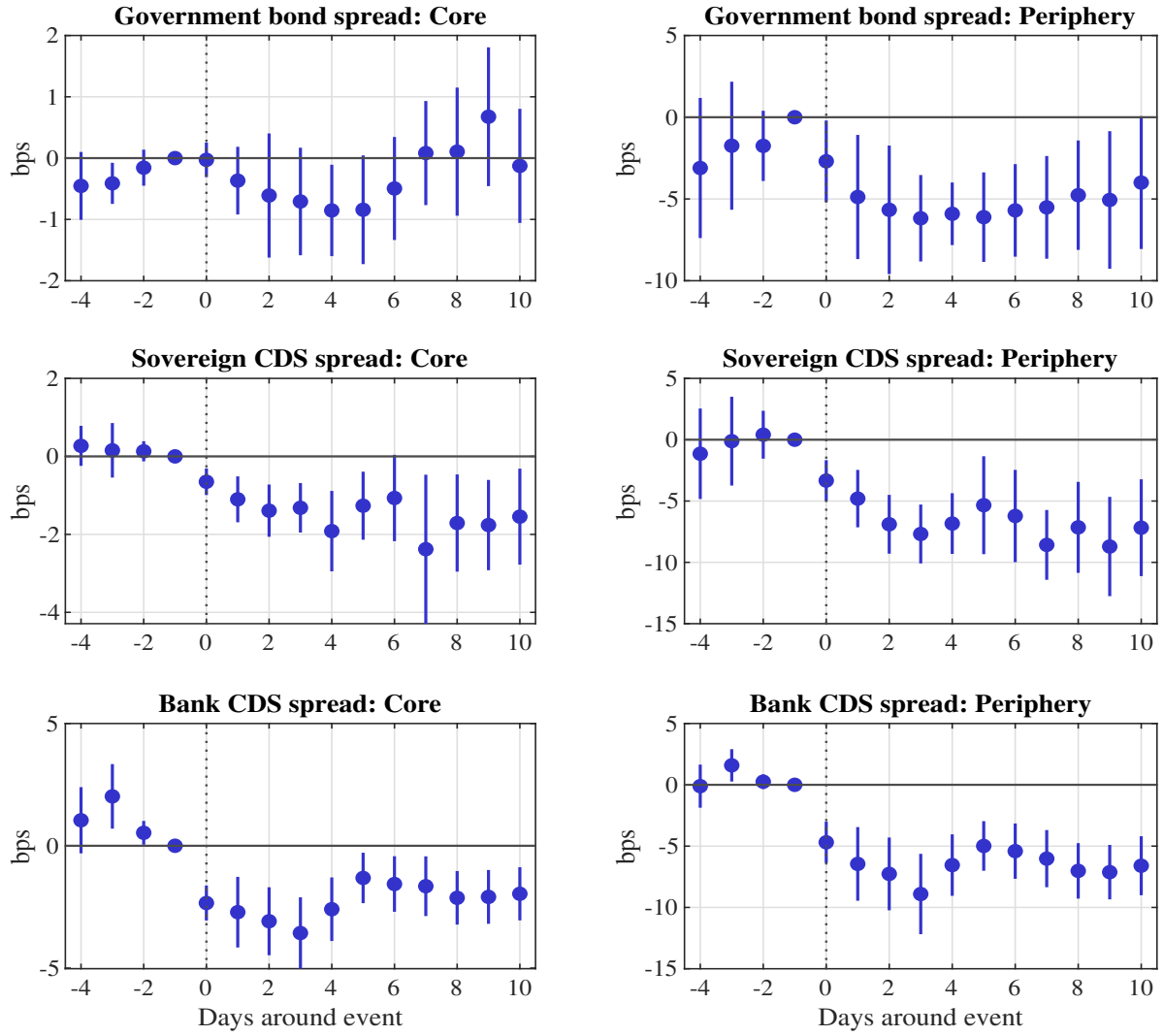


Figure B.7: Pre-trend sensitivity check for transmission to government bond market and bank default risk: Results from estimating equation (10) on the full sample. The top row reports 5-year government bond yield spreads relative to maturity-matched OIS. The mid row reports 5-year sovereign CDS spreads. Core countries are Germany and France; periphery countries are Italy and Spain. The bottom row reports 5-year bank CDS spreads, split based on bank location. Dots represent estimated coefficients and vertical lines denote 90% confidence intervals based on heteroskedasticity- and autocorrelation-consistent (Newey-West) standard errors. The horizontal axis reports event time in days relative to the policy announcements, with $t = 0$ marking the announcement dates.

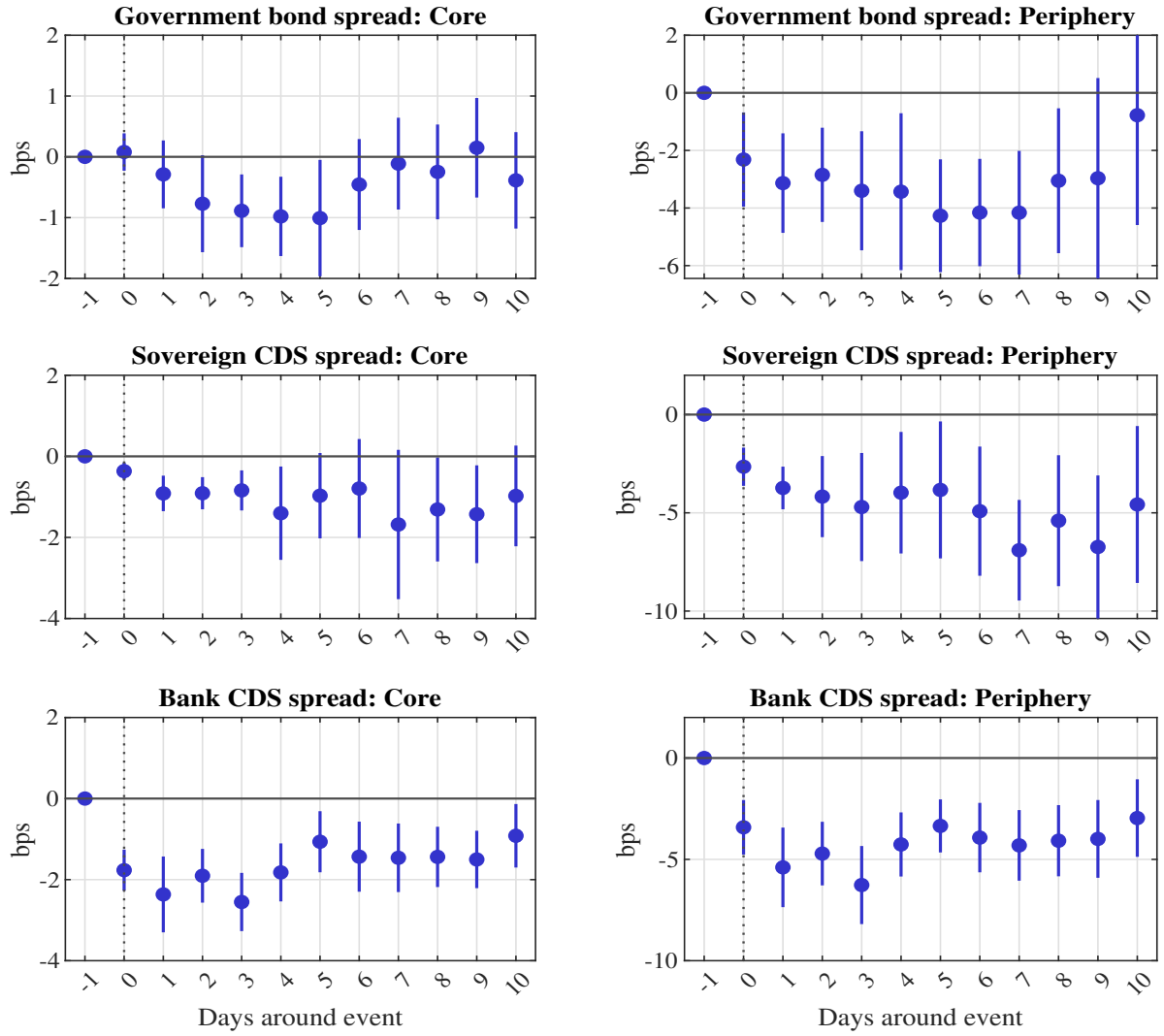


Figure B.8: Orthogonalized CPS sensitivity check for transmission to government bond market and bank default risk: Results from re-estimating equation (10) using collateral policy surprises orthogonalized with respect to the potential predictors included in specification (3) of Table 1. Following Bauer and Swanson (2023), the orthogonalized surprise series is constructed as the residual from regressing the baseline collateral policy surprises on these predictors. The top row reports 5-year government bond yield spreads relative to maturity-matched OIS. The mid row reports 5-year sovereign CDS spreads. Core countries are Germany and France; periphery countries are Italy and Spain. The bottom row reports 5-year bank CDS spreads, split based on bank location. Dots represent estimated coefficients and vertical lines denote 90% confidence intervals based on heteroskedasticity- and autocorrelation-consistent (Newey-West) standard errors. The horizontal axis reports event time in days relative to the policy announcements, with $t = 0$ marking the announcement dates.

B.4 Exchange Rate and CIP

In the main text, we demonstrate that collateral policy does not affect the EUR/USD spot exchange rate. However, this does not imply that cross-border funding conditions remain unaffected. Recent work emphasizes the global risk-taking channel and the role of the U.S. dollar in shaping international financial conditions (Bruno and Shin, 2015; Rey, 2015; Miranda-Agrippino and Ricco, 2021). In this framework, monetary policy affects global financial conditions by altering leverage, risk-taking, and cross-border funding structures rather than solely through the traditional trade or exchange rate channel. While much of this literature focuses on U.S. monetary policy, recent evidence suggests that ECB policy shocks can also generate meaningful international financial spillovers, albeit smaller in magnitude (Miranda-Agrippino and Nenova, 2022; Corsetti et al., 2024).

Against this background, collateral policy surprises may also operate through a funding channel. By relaxing euro funding constraints for banks active in international markets, collateral policy reduces the relative cost of EUR funding. If EUR funding becomes cheaper relative to USD funding, banks may increasingly borrow in EUR and obtain USD via FX swaps rather than borrowing USD directly. This shift in funding structure increases pressure in FX swap markets and can generate deviations from covered interest parity (CIP), reflected in a more negative EUR/USD basis, even in the absence of spot exchange rate movements. To shed light on this mechanism, we compute the CIP-implied forward exchange rate

$$F_{t,T}^{CIP} = S_t \times \frac{1 + r_{t,T}^{USD}}{1 + r_{t,T}^{EUR}}, \quad (\text{B.1})$$

where S_t denotes the spot USD/EUR exchange rate, $r_{t,T}^{USD}$ the USD money market rate, and $r_{t,T}^{EUR}$ the euro-area money market rate for maturity T . The CIP is defined as the deviation between the observed forward rate and the CIP-implied forward rate,

$$\text{CIP}_{t,T} = F_{t,T}^{obs} - F_{t,T}^{CIP}. \quad (\text{B.2})$$

We then estimate (10) for the spot rate, the 1-year forward rate, and the 1-year CIP; results are shown in Figure B.9. Collateral policy shocks do not significantly affect the EUR/USD spot or forward rate. In contrast, we observe a sizeable and persistent decline in the 1-year CIP. This indicates that collateral policy primarily operates through international funding markets rather than through exchange rate adjustment.

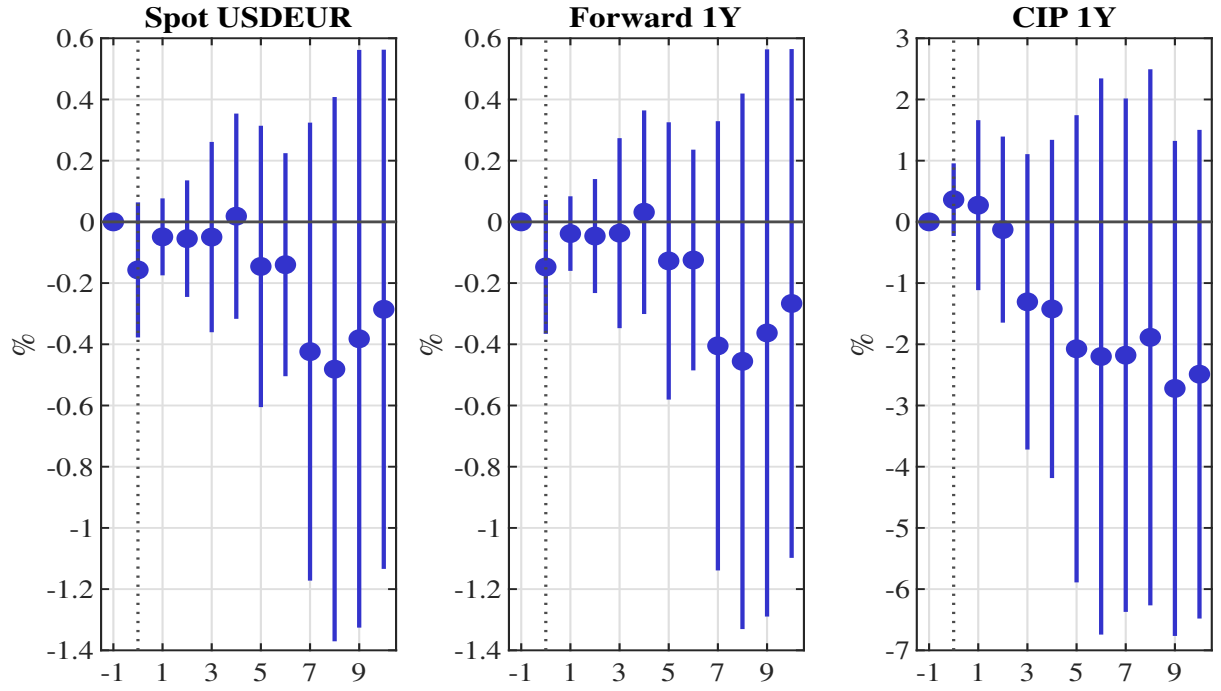


Figure B.9: Collateral Policy Surprises; Spot, Forward Rate and Covered Interest Rate Parity: Results from estimating (10) for deviations from covered interest rate parity (CIP) at the 1-year maturity. The CIP-implied forward exchange rate is computed as

$$F_{t,T}^{CIP} = S_t \times \frac{1 + r_{t,T}^{USD}}{1 + r_{t,T}^{EUR}},$$

where S_t denotes the spot USD/EUR exchange rate, $r_{t,T}^{USD}$ the USD money market rate, and $r_{t,T}^{EUR}$ the euro-area money market rate for maturity T . The CIP is defined as the deviation between the observed forward rate and the CIP-implied forward rate,

$$CIP_{t,T} = F_{t,T}^{obs} - F_{t,T}^{CIP}.$$

Blue shaded areas denote 68% and 90% confidence intervals computed from heteroskedasticity-consistent (White) standard errors. Horizon in days.

B.5 Monthly Local Projections

We complement the daily high-frequency estimates with monthly panel local projections to assess the persistence of the effects of collateral policy surprises. To do so, we aggregate daily outcome variables to monthly frequency using the last available trading-day value of each month, while daily collateral policy surprises are aggregated by summing all surprises within the month. We first estimate, for each financial market outcome y_m (month m),

$$y_{m+h} = \beta_h s_m + \sum_{k=1}^{12} \gamma'_k X_{m-k} + \alpha + \varepsilon_{m+h}, \quad h = 0, \dots, 6, \quad (\text{B.3})$$

where s_m is the monthly aggregated collateral policy surprise, X_{m-k} collects twelve monthly lags of common financial market variable controls and α is a constant. Standard errors are HAC.

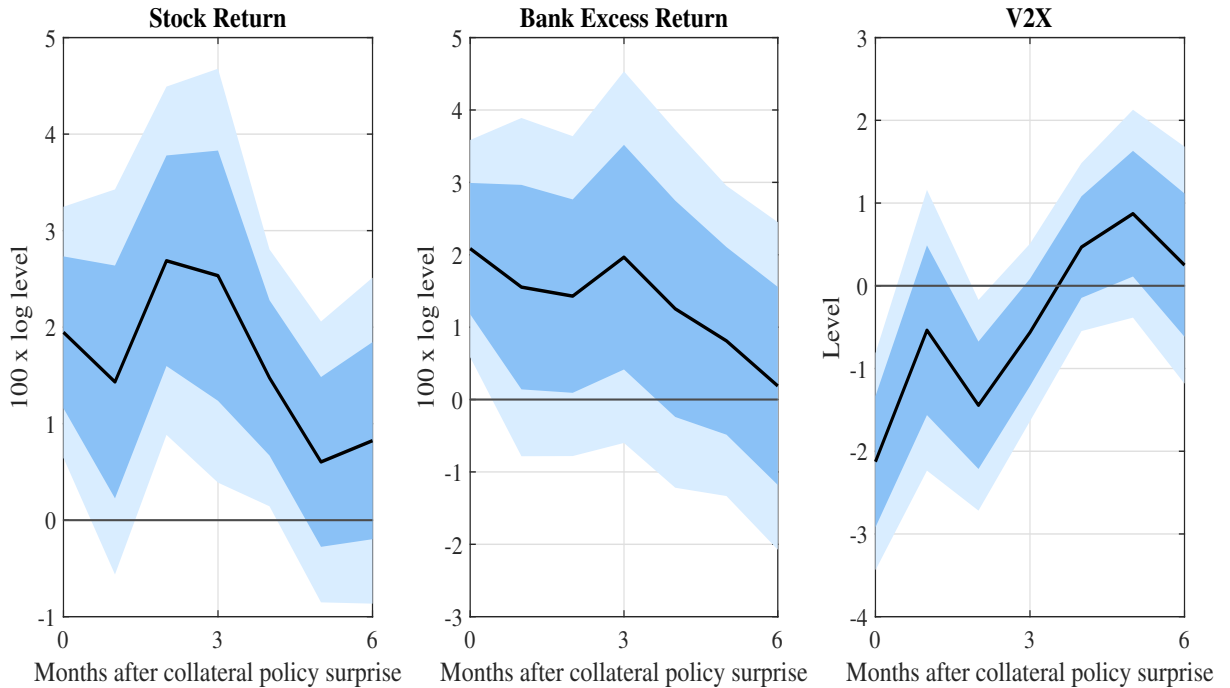


Figure B.10: Monthly Panel Local Projections: Financial Market Results Results from estimating (B.3) at a monthly frequency for financial market outcomes. The dependent variables are the EuroSTOXX50 return, the excess return of EuroSTOXX Bank Index over the overall index, and the implied volatility of EuroSTOXX50 options (V2X). Daily outcomes are aggregated to monthly frequency using the last available trading-day value of the month; the collateral policy surprise is aggregated by summing all surprises within the month. Regressions include twelve monthly lags of common financial market controls. Blue shaded areas denote 68% and 90% confidence intervals computed from heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. Horizon in months.

We then estimate (B.3) at a monthly frequency first for the financial market outcomes. Figure B.10 presents the results. Following a collateral policy surprise, the EuroSTOXX50 return and the bank excess return both rise on impact and remain positive through month 3, before declining towards zero over the remainder of the six-month horizon; The response of implied volatility (V2X) moves in the opposite direction, falling sharply on impact and remaining significantly negative for almost one quarter. Taken together, these patterns are consistent with

the results found at the daily frequency. Collateral easing improves financial conditions, and the effects vanish within about half a year.

For the government bond market analysis, we add a Core/Periphery interaction to (B.3), everything else equal. For each outcome $y_{i,m}$ (country or bank i , month m), we estimate

$$y_{i,m+h} = \beta_h^{Core} (Core_i \times s_m) + \beta_h^{Periphery} (Periphery_i \times s_m) + \sum_{k=1}^{12} \gamma'_k X_{m-k} + \alpha_i + \varepsilon_{i,m+h}, \quad (\text{B.4})$$

where $h = 0, \dots, 6$ and $Core_i$ and $Periphery_i$ are mutually exclusive group dummies, α_i denotes country or bank fixed effects, and s_m and X_{m-k} are defined as above.

Turning to the government bond market, Figure B.11 shows the results for the 5-year sovereign bond spread over OIS, the 5-year sovereign CDS spread, and bank CDS spreads. Periphery government bond and sovereign CDS spreads decline sharply and significantly on impact and gradually revert towards zero over the six-month horizon. Core spreads move by much less, and the confidence bands include zero at most horizons. A similar picture also emerges for bank CDS spreads. This indicates that the core-periphery asymmetry documented at daily frequency is also present at the monthly frequency.

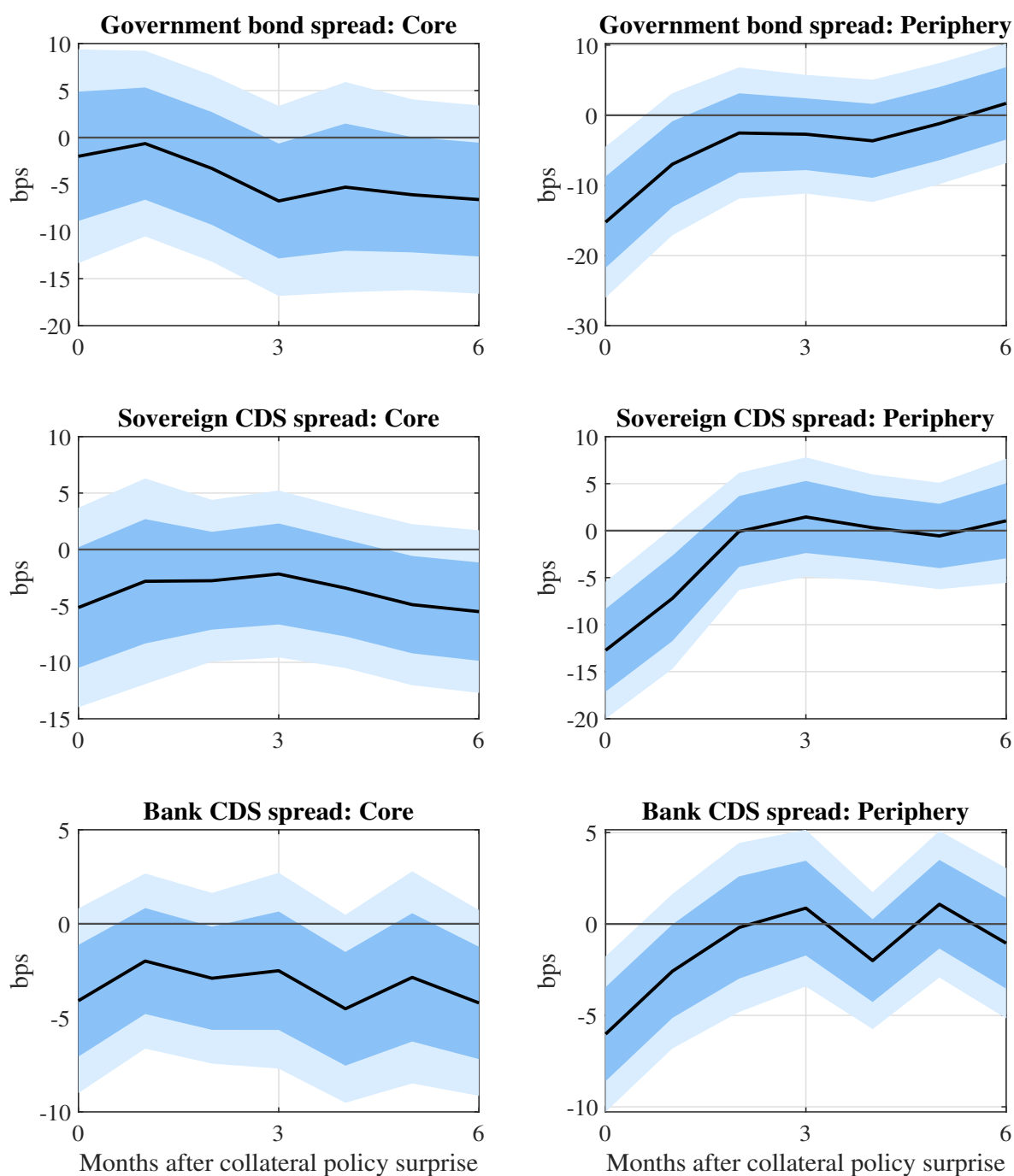


Figure B.11: Monthly Panel Local Projections: Government Bond Market Results: Results from estimating (B.4) at a monthly frequency for the 5-year government bond spread over OIS (top row), the 5-year sovereign CDS spread (middle row), and bank CDS spreads (bottom row), separately for Core and Periphery (columns). Daily outcomes are aggregated to monthly frequency using the last available trading-day value of the month; the collateral policy surprise is aggregated by summing all surprises within the month. Regressions include twelve monthly lags of common financial market variables controls and country/bank fixed effects. Blue shaded areas denote 68% and 90% confidence intervals computed from heteroskedasticity-and autocorrelation-consistent (HAC) standard errors. Horizon in months.